Supply-Side Responses in School Choice

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Abstract

This paper examines the competitive effects of the Indiana Choice Scholarship Program on public and participating private schools. Using a difference-in-differences design, we find that high exposure public schools witnessed increases in quality, driven by improvements in math. Initially poor performing public schools drive our results, suggesting that the public school quality gap shrank because of the program. Average quality at participating private schools fell following the program's adoption. We also present suggestive evidence that high exposure participating private schools saw larger declines in quality. Policymakers should consider these indirect effects to understand vouchers' total impact on educational outcomes.

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I Introduction

School choice programs have become a popular tool to eliminate inequities in access to schooling. Private school vouchers have drawn increasing attention in this effort. In the last 20 years, the number of state-funded voucher programs has increased five-fold, from 5 in 2000 to 25 in 2024. Furthermore, the scale of these programs has grown significantly over time. The first U.S. voucher program, Milwaukee Parental Choice, featured an enrollment limit of 1% of the public school population when it launched in 1991. Today, the average voucher program has no enrollment cap, and around 26% of families qualify to participate (EdChoice, 2024). Moreover, in states that have these programs, nearly 1 in 10 private school students now use a voucher to attend (EdChoice, 2024; National Center for Education Statistics, 2019).

The introduction and expansion of voucher programs could impact education outcomes through two main channels. First, voucher programs could expand access to higher quality schools for qualifying students. Second, voucher programs could alter school quality by changing competition over enrollment. Most of the existing literature evaluating the effectiveness of these programs focuses on the fist channel through the use of small-scale randomized control trials (RCTs) to compare the outcomes of those offered a voucher to those in the control group for a small subset of the total student population (Mayer et al., 2002; Howell et al., 2002; Wolf et al., 2010; Witte et al., 2014; Abdulkadiroğlu et al., 2018).[1] While these RCTs provide useful estimates of the average effect of being offered a voucher, understanding the competitive effects of these programs, and how they might differ across the public and private school sectors, is essential for understanding how such programs impact educational outcomes for students not directly participating in the program.

In this paper, we quantify the competitive effect of private school voucher programs by examining changes in school quality following the adoption of one of the largest voucher programs in the United States. Our context centers around the Indiana Choice Scholarship Program (ICSP), which was initially adopted in 2011 and expanded in 2013. We begin with student-level testing data that covers all students in the state between the 2005-2006 and 2017-2018 academic years (AY). We use these data to construct school-level measures of quality by estimating value-added for both public and private schools. We then identify schools facing greater exposure to the voucher policy by calculating the radial distances between each public and private choice school within the state. Specifically, we distinguish high exposure public schools as those that face increased competition because they are located within five miles of a private school that eventually accepts voucher students (hereon called private choice schools). Private choice schools are said to face high exposure to the policy if they are

[1]See Epple et al. (2017) and Rouse and Barrow (2009) for excellent reviews on the topic.

in the top tercile of the distribution of the number of public schools within five miles.[2] The resulting data sets track school quality for those in our high exposure and control groups, both before and after the implementation of the voucher program.

Using these datasets, we estimate the causal effects of the implementation of ICSP on schools using a standard difference-in-differences model. Specifically, we compare the change in school value-added in the years before and after the implementation of ICSP for schools facing high exposure to the policy versus those in the control group. We separate our analyses for public and private choice schools. Beginning with the public schools, we find that, on average, schools facing the threat of voucher competition saw a statistically significant increase of 0.020 of a standard deviation (s.d.) in their overall school value-added, with the improvements being largely driven by increases in math value-value added. These improvements in value-added varied within the high exposure group. Public schools facing the threat of competition and an above-median share of students qualifying for free or reduced-price lunch witnessed the largest improvements in school quality. We might expect these schools to have a greater response since a larger share of their students automatically qualify for a voucher. Specifically, these schools saw increases of 0.027 s.d. in overall school VA, 0.044 s.d. in math VA, and 0.018 s.d. in reading VA. We also find that after the adoption of ICSP, high exposure public schools saw increases in their attendance and no changes in the percent of students ever suspended or expelled. These findings suggest that in response to ICSP, schools increased quality in ways that improved outcomes beyond test scores.

We further explore the competitive impacts of ICSP on public-school quality by disaggregating the results by several baseline characteristics. Specifically, we examine differential impacts based on baseline enrollment, overall school value-added, income of the census block group where the school is located, and the quality of the nearest private choice school competitor. Both smaller public schools and schools in lower-income neighborhoods may be more sensitive to changes in enrollment and might increase quality to avoid risking closure. Similarly, public schools that were initially poor performing or were of lower quality that the nearest private choice school competitor may face additional pressure as the voucher program allows parents to exercise an additional form of choice. While we find no evidence of heterogeneous results across enrollment or household income, there

[2] The definition of "high exposure" shifts between public and private choice schools because while about 60% of the public schools have a private choice school within five miles, 98% of private choice schools have a public school within five miles. The use of a radial distance to distinguish between treated and control groups has been used in several contexts including understanding the role of traffic conditions on infant health outcomes (Currie and Walker, 2011), the impacts of the introduction of charter schools on traditional public schools (Cordes, 2018), and the effects of fracking on infant health through drinking water quality (Hill and Ma, 2022). In Section III.B, we discuss alternative methods for distinguishing high exposure schools.

are significant differences in outcomes based on initial quality. We find that high exposure public schools with an above-median baseline school value-added saw almost no changes in our outcomes of interest. This result suggests that initially poor-performing public schools facing the potential threat of competition drive the changes we see in quality. Together, these results lead us to conclude that the gap in public-school quality shrank following the implementation of ICSP. We also find that high exposure public schools whose closest eventual private choice school is of higher initial quality see the largest improvements across all of our outcome variables of interest, bolstering the claim that public schools are responding to competitive threats.

We also employ an event-study specification that allows us to examine whether the adoption and expansion of ICSP had differential impacts on public school quality. We find that the adoption of the policy did not elicit differential changes across high exposure and control public schools in our outcome measures of interest. Instead, increases in school quality among high exposure public schools are seen only after the program's expansion. This result indicates that despite facing potential enrollment losses when the program was adopted, public schools only responded once there was a threat that a majority of their students could leave.

To understand how high exposure public schools increase quality, as measured by VA, we combine a school-level dataset on available teachers between the 2010-2011 and 2017-2018 academic years with the National Center for Education Statistics' Common Core of Data on Indiana public schools. We do not find strong evidence that following the implementation of ICSP, high exposure public schools saw changes in their student-teacher ratios. However, high exposure public schools saw an increases of 0.6 in number of teachers with a graduate degree, 1.41 in number of teachers with a high-quality certification and a 2 p.p. increase in the teacher retention rate when compared to the set of control schools. These results suggest that changes in teacher characteristics may be an important response public schools have to increased competition.

Given our results, we pay particular attention to the possibility that changes in the composition of students could generate our public school findings. To address this concern, we first want to highlight that our public school value-added measures are estimated off the set of students that never use a voucher and do not qualify for a 90% voucher. We then document the extent to which student sorting occurs after the adoption of ICSP. We find that high exposure public schools see a decline of 2.9 percentage points (p.p.) in the share of White students and a rise of 2.6 p.p. in the share of Hispanic students after the policy was adopted. To understand whether these demographic changes drive our results, we run a difference-in-differences specification using predicted value-added. We find that based only on changes in observable characteristics, high exposure public schools were predicted to see declines in their school value-added. We take these results as evidence that the improvements in school quality are not due to student sorting.[3] Additionally, we use information on student-class links to create a sample of students in the public schools that did not have classes with ever voucher students. We re-estimate school value-added with this sample under the assumption that school quality calculated with this group of students would be less impacted by potential peer effects. Our difference-in-differences results using this sample continue to show that high exposure public schools saw meaningful increases in school quality following the implementation of ICSP, further bolstering our claim that composition of students does not drive our results.

Our public school results are robust to model specification choices and the adoption of other policy interventions that could threaten the validity of our findings. First, in our event-study specifications, high exposure public schools and those in the control group appear to have similar trends in school value-added in all years prior to the program, suggesting that our results are not driven by differential trends between the two groups of public schools. We also show that our results are robust to placebo adoption years and find that changes in school quality occurred only after the expansion of the voucher policy, further bolstering our conclusions that pre-trends do not drive our results. As an additional validity check, we employ the event-study sensitivity analysis proposed in Rambachan and Roth (2023). We find that our treatment effects are robust to allowing for violations of parallel trends up to the max violation in the pre-trend period. To address the concern that high exposure public schools may be concentrated in a small number of urban districts, we run our results dropping each county in Indiana. The analysis produces similar results to the entire state sample. Lastly, we argue that no meaningful policy changes were adopted that would have differentially impacted our two sets of schools and influenced our findings.

We then move to the results for private choice schools. How might the response of private choice schools differ from public schools? Voucher programs cause an increase in demand for private choice schools by decreasing the price for students to attend. As a result, private choice schools face a larger market of potential enrollees. That same price reduction causes public schools to face more competition as students who may not have been able to afford private schools may now leave the public school system. Thus, private choice schools and public schools may respond differently in their efforts to improve school quality due to the different effect on the competitive pressure they experience resulting from the introduction of vouchers.

We find that private choice schools see declines in average quality on all dimensions during the first year of the voucher program. In our difference-in-differences specification, we compare private

[3] We address concerns over non-random student sorting on unobservable characteristics by highlighting the advantages of our value-added estimates since they control for prior achievement. Assuming that prior achievement fully proxies for inputs that affect a student's achievement prior to using the voucher and those inputs are correlated with a student's likelihood of using a voucher, we can mitigate the concerns of this type of sorting.

choice schools surrounded by many public schools to those with fewer options to attract students. While statistically insignificant, we find consistent evidence that high exposure private choice schools saw larger decreases in school quality compared to the control group. Importantly, these findings were not the result of negative selection into the program. Our heterogeneity analysis shows that the declines in school quality were driven by high exposure private choice schools with either a non-"A" grade prior to ICSP or above median baseline school value-added.

We use the Private School Universe Survey to understand to what extent private choice schools alter their school inputs during our sample period. Following the adoption of ICSP, high exposure private choice schools see a statistically significant increase of 0.83 (off a base mean of 14.22) in their student-teacher ratios. We also find suggestive evidence that control choice schools increase their instructional time to catch up to high exposure choice schools once the program is adopted.

Our paper contributes to the growing economics literature on school choice programs. Many papers examining private school vouchers focus on the direct impact of these policies on the educational outcomes of students offered to participate. One set of papers examines whether participating students experience test score gains (Rouse, 1998; Mayer et al., 2002; Howell et al., 2002; Witte et al., 2014; Wolf et al., 2010; Abdulkadiroğlu et al., 2018; Waddington and Berends, 2018), and Chingos and Peterson (2015) focuses on the longer-term educational impacts including high school graduation and college enrollment. Our paper complements this prior work by demonstrating that the competitive effects of voucher policies impacts the educational outcomes of students not participating in the program. Specifically, we show that as ICSP is implemented, both students remaining in the public school system and those continuing in the private choice schools experience changes in school quality. By establishing these competitive effects of ICSP on public and private choice schools, we can better understand the total effect of voucher policies as they are adopted and expanded.[4]

A large body of work evaluates the supply-side responses to school choice programs. Many of these papers focus on the public school response to the introduction of charter schools (Imberman, 2011b; Gilraine et al., 2021; Cohodes and Parham, 2021; Slungaard Mumma, 2022; Figlio et al., 2024).[5] We give two reasons why understanding voucher programs' specific effects are important. First, current policy discussions often center around the adoption and expansion of voucher policies in particular. In the last five years, policymakers from Oklahoma, Nevada, Texas, Missouri, and Florida have made public announcements supporting the introduction or expansion of voucher policies. Second, the competitive effects from the introduction of charter schools are likely distinct from private school voucher programs. Charter schools mainly draw their enrollments from neigh-

[4] More broadly, this paper contributes to the literature that calls attention to the limitations of randomized control trials (Lise et al., 2004; Heckman, 1991; Deaton and Cartwright, 2018; Al-Ubaydli et al., 2017).

[5] See Epple et al. (2016) for an excellent review of the effects of charters schools on public school performance.

boring public schools (Chakrabarti and Roy, 2016), suggesting that the competitive effects of their introduction are concentrated on traditional public schools rather than private schools. Standard voucher programs, such as ICSP, induce competition across both public and private schools.

Of similar interest is the distinction between standard voucher programs, like ICSP, and Education Savings Accounts (ESAs). ESAs allow families to use funds to cover expenses beyond tuition and fees to include items like the purchase of online learning programs and private tutoring. Broad-eligibility ESAs are only available in a few states as these programs are often centered around students with disabilities. The main distinction between (broad based) ESAs and standard voucher programs is the types of schooling they promote. Standard voucher programs only incentivize the move to participating private schools, while ESAs are setup up in such a way that also promotes homeschooling. This difference changes the type of competition private schools face. However, understanding how standard voucher programs impact school quality is informative for states considering the adoption of an ESA program, since they are similar.

The most similar work to ours examines the response of public schools to the voucher programs in Milwaukee, Ohio, and Florida (Hoxby, 2003; Figlio and Rouse, 2006; Chakrabarti, 2008, 2013; Rouse et al., 2013; Chiang, 2009; Greene and Marsh, 2009; Figlio and Hart, 2014).[6] Overall, these studies investigate the introduction of school voucher programs and find modest positive effects on public school performance. Our context has attractive empirical properties that allow us to avoid some identification issues present within the literature.[7] Figlio et al. (2023) also studies the competitive effects of a large-scale voucher program by leveraging the Florida Tax Credit Scholarship's (FTC) growth from 2003 to 2018. The authors use variation in the growth of the program and pre-policy levels of local competition to estimate the intensive marginal effects of increased competition on public school performance. They find that students in public schools that faced a higher initial level

[6]There are several studies examining the specific effect of voucher policies on schools in countries outside of the United States (Sandström and Bergström, 2005; Hsieh and Urquiola, 2006; Neilson, 2021; Böhlmark and Lindahl, 2015; Muralidharan and Sundararaman, 2015). These papers are similar to ours in that they study voucher programs with broad-based eligibility. However, U.S. private school enrollment is a much smaller share of K-12 education as compared to these international contexts, suggesting the competitive threat of ICSP is likely different. Furthermore, we answer different questions than those in the cited text. Hsieh and Urquiola (2006) and Böhlmark and Lindahl (2015) focus on the average aggregate effects of voucher programs, while Sandström and Bergström (2005) and Neilson (2021) analyzes the impacts of voucher programs only on one sector of the education market. Muralidharan and Sundararaman (2015) provide an experimental evaluation of the total effect of vouchers in the Indian state of Andhra Pradesh, but the scale of the program is much smaller than ICSP.

[7]Figlio and Hart (2014) mentions that several papers rely on changes in the degree of private school supply for identification, which may be endogenous to public school performance. Other papers identify the effects of voucher programs by leveraging policies that allow students to qualify if their school receives a repeat "F" grade. However, these papers cannot disentangle the effects of school vouchers from the performance effects of accountability pressure.

of competitive pressure saw greater gains in test scores as the program matured.

We build on their results in several ways, beginning with our identification strategy. Rather than rely on incremental changes in realized voucher enrollment, our results are estimated off legislated changes in eligibility. Understanding the effects based on this dimension may be of particular interest as policymakers can set the limits for eligibility and voucher amount and cannot directly control the number of students participating. Furthermore, their results combine the effects of FTC scaling up and maturing over time. Our analysis centers around the first seven years after ICSP was adopted, where maturation effects are less apparent. Our heterogeneity analysis (focused on school attributes) also better identifies which public schools respond to the competitive threat of voucher programs. We show that while distance to nearest competitor matters, the largest improvements came from schools who had an initially high share of students that automatically qualified for the program, were below the median in baseline school value-added and whose nearest competitor was of higher baseline quality. Examining where voucher programs create the largest competitive threat is important for policymakers to understand which schools will be impacted as these programs continue to be adopted and expanded. Additionally, our ability to explore heterogeneity by nearest competitor quality is unique because of the data needed from private schools.

We also examine changes in the quality of private choice schools, which is critical for examining voucher programs' total impact. To the best of our knowledge, we are the first to examine changes in private school quality in response to a voucher program within the United States.[8] Student-level testing data for those in private schools is rare, simply because many states don't require private schools to administer standardized tests. Even within states that have larger voucher programs, testing is often only required for students participating in the program (e.g., Florida, Ohio and Louisiana). This fact suggests that Indiana is one of the few contexts in the United States where we can learn about changes in private school quality due to these programs.

Egalite and Catt (2020) also study the competitive effects of ICSP but focus the analysis on remaining public school students. The authors use a similar estimation strategy and find that low-income students remaining in the public school system saw small increases in test scores. We add to their results by focusing our analysis at the school-level. Our data allows for additional analyses that are important for broadening our understanding of the competitive effects of ICSP. Specifically, we are able to include a discussion of mechanisms, isolate where competitive effects are strongest

[8] Private school responses to voucher programs in the United States is an understudied area. Some papers have studied the effects of these policies on private school enrollment, finances, and school inputs; however, the question of whether and to what extent schools alter quality is still an open question (Hungerman and Rinz, 2016; Hungerman et al., 2019; Rinz, 2015).

through our heterogeneity analysis and include the analysis of changes in private school quality.[9]

The remainder of this paper is organized as follows. In Section II, we provide background information on the Indiana Choice Scholarship Program. Section III summarizes the data used in this paper and describes our constructed measures of school quality and exposure to the policy. Section IV describes the reduced-form empirical strategy and lays out the regression specifications. Section V contains the main results, which include our heterogeneity analysis, discussion on student sorting, our validity checks, and a discussion on possible mechanisms. Section VI contains the results for private choice schools. Section VII offers conclusions from this research.

II The Indiana Choice Scholarship Program

The Indiana Choice Scholarship Program (ICSP) is one of the most expansive single voucher programs in the United States in terms of both participation (36,290 participants) and eligibility (over 79% of families with children are eligible). In 2018, Indiana ranked sixth in terms of percentage of current educational expenditures spent on voucher programs. Initially, the program capped participation at 7,500 and 15,000 students for the 2011-2012 and 2012-2013 AYs, respectively. The expansion of ICSP at the start of the 2013-2014 AY eliminated participation caps and expanded the paths available to students to qualify for a voucher. Since the expansion, a student can participate in ICSP if they meet the income requirements and qualify under one of eight eligibility tracks.[10]

Income eligibility for vouchers is based on household size and is set as a percentage of the amount to qualify for the Federal Free or Reduced-Price Lunch (FRPL) Program. Students at or below the threshold for FRPL are eligible for a voucher of value up to 90% of per-pupil state funding, while students at or below 300% of the threshold for FRPL are eligible for a voucher of value up to 50% of per-pupil state funding (Indiana Department of Education, 2021a). The actual voucher amount equals the minimum of school tuition and fees or the qualified voucher amount. During the 2017-2018 school year, the average voucher amount for students in grades 1-8 was \$4,818 for students qualifying for the 90% voucher and \$2,862 for those receiving the 50% voucher ($\leq$ 50% of per-pupil public spending) (Indiana Department of Education, 2019).

For a student to receive a voucher, they must apply and be accepted into a participating private choice school. The choice scholarship application is then completed by a parent (or legal guardian) and submitted by the private school. If a student is awarded a voucher, that money goes directly

[9]We are also able to leverage additional years of pre-policy data. While Egalite and Catt (2020) also use a difference-in-differences design, they only have access to one year of data prior to the adoption of ICSP; therefore, the authors are unable to include any event-study specifications.

[10]Information on available tracks can be found on the IDOE website (Indiana Department of Education, 2021b).

to the school, and only an award letter detailing the approved amount of the voucher is given to parents. The distribution of funding to schools rather than households distinguishes ICSP from tax-credit voucher programs or educational savings accounts. ICSP vouchers are meant to cover tuition and fees at eligible private schools; however, these schools are allowed to charge additional tuition above the voucher amount so long as they are the same charges non-choice-eligible students pay. Around 45% of students participating in the voucher program paid zero dollars in out of pocket expenses to attend the private choice school during our sample time period.

For a private school to participate in the voucher program, the school must submit an application for the upcoming academic year to the Indiana Department of Education between March 1 and September 1. The department then approves the application within 15 days, so long as the school is eligible. Eligible schools include nonpublic schools that are willing to accept eligible students, are accredited by the state or by an accreditation agency approved by the state and administers the statewide assessment (Indiana Code 20-51-4-7).

The inclusion of both low- and modest-income families makes ICSP unique. The income eligibility threshold for a household of four in the 2017-2018 academic year in Indiana was about 1.3 times that of the Florida voucher program (Fla. Stat. § 1002.394); 1.25 times higher than that of the programs in Milwaukee (Wis. Stat. §§ 119.23 and 235), and Racine, (Wis. Stat. § 118.60); about 1.5 times higher than the program in New Orleans (La. Rev. Stat. §§ 17:4011 through 4025) and 2 times higher than the program in Washington, D.C. (DC ST § 38-1853). This higher income threshold places additional pressure on the public schools of Indiana. Over 79% of public-school students qualify for a voucher, and participation is not capped at a percentage of public-school enrollment as seen in other voucher programs, suggesting that Indiana is a context where we might expect to see larger impacts on school quality.

Table 1 shows some additional descriptive statistics about the growth of the Indiana Choice Scholarship Program during our sample time period. We find that both eligibility and participation were broad following the expansion of the program. Importantly, while participation remained around 3% for all K-12 students at the end of our time frame, the competitive threat faced by public schools was large. This finding suggests that there are barriers to take-up. However, participation rates may be low because of the increases in public school quality we document. The competitive threat of private school voucher programs incentivizes public schools to increase quality to prevent potential reductions in enrollment and therefore participation in the program. Second, it is likely that eligibility (along with having a nearby competitor) could be a driving force behind the competitive pressure faced by public schools. Because of the relatively high income limits of ICSP, a large majority of public school students would automatically qualify for the voucher if families wanted to participate. Furthermore, a large share of those eligible students would have zero dollars to pay in out-of-pocket expenses to attend a private choice school. We believe that these facts, together, are why we find stronger results for schools with high shares of pre-policy FRPL enrollment as shown in the following section.

III Data

The data for this project come from the Indiana Department of Education (IDOE) through a data agreement with the Center of Research on Educational Opportunity (CREO) at the University of Notre Dame. The IDOE-CREO database contains student-level data with information on the membership, test scores, voucher take-up, and demographics of all students enrolled in a public, accredited private, or charter school in Indiana. We focus on public and private school students in this paper. The database covers the 2005-2006 through 2017-2018 AY. We focus on students in schools that serve anyone in grades 3-8. Standardized testing is consistent between these grades and is required in both public and private schools in order to remain accredited (Indiana Code §20-32-5-17), which allows for a consistent sample across the sample years. Our dataset is advantageous because it includes testing data for private schools prior to the adoption of ICSP and covers both voucher and non-voucher students. Many programs require participating private schools to administer state exams once they accept voucher students, but this means testing data only exists post-adoption and often only for students using a voucher. Indiana private schools had the incentive to be accredited before ICSP because it was required if a school wanted to participate in the Indiana Athletic Association (Indiana Athletic Association, 2021).

Demographic information in the IDOE-CREO database varies depending on whether a student attends a public or accredited private school. While we have testing data on all accredited private schools our private school sample will focus exclusively on participating schools (private choice schools). For all students, we have information on race, age, date of birth, free or reduced-price lunch status, Section 504 status, zoned school district, and standardized testing accommodations. For students attending either public or accredited private schools, we have information on whether a student would qualify for a 90% voucher as it is the same cutoff for free/reduced-price lunch. We have additional information on students that use a voucher to attend a private choice school. Specifically, we also have information on these students' home addresses, the tuition they are charged, their voucher status (50% or 90%), and the amount of the voucher they receive.

We also have access to school directories that outline basic information about the schools in Indiana. This includes data on the opening and closing (if applicable) dates, addresses, school type (public, accredited private, or charter), and lowest/highest grades offered. We construct school-level test scores and demographic information by aggregating individual-level data from students attending each school. Schools must have non-missing test score data for each of the academic years between 2005 and 2017 to be included in the sample. Therefore, none of the private choice or public schools in our sample have incomplete data. After the estimation of school value-added, 1,269 public elementary and middle schools and 179 choice schools remain.[11] We show that our results do not hinge on this sample restriction in Appendix Table A11.

We create two other school-level measures for our analysis: school value-added, which is used as our proxy of school quality, and our measure of high exposure to the policy, which is used to distinguish schools in our treatment and control groups. The following sections explain how those measures were created.

III.A School Value-Added Estimates

School value-added (VA) is a measure of a school's contribution in a given year to students' test scores. We use it as our proxy for school quality, with the assumption that this measure captures how much a school increases students' achievement, controlling for all other relevant variables. This measure of school quality is meant to capture schools' inputs such as teacher quality, infrastructure, school environment, and any other school-specific characteristic that improves student achievement, measured as the average test score.[12]

To calculate school VA, we run the following OLS regression:

$$testscore_{ist} = \alpha + \sum_{g} \gamma_g testscore_{ist-1} + \sum_{g} \lambda_g testscore^2_{ist-1} + \mathbf{X}_i'\delta + \beta_{st} + \epsilon_{ist} \tag{1}$$

where $testscore_{ist}$ is the test score for a student i, at school s in year t. Students in the third through eighth grade take both a Math and an English language arts exam each year; thus, we have school VA estimates for each subject as well as for the average of both scores. These scores are standardized within grade and year so that estimates can be interpreted as standard deviations.[13] $testscore_{ist-1}$ is the student's test score from the previous academic year and is constructed in the same manner as

[11]This restriction necessarily means the set of schools in our sample is positively selected. We discuss entry into and exit from the educational market in Appendix Section B1. We can make a few comparisons across private schools in and out of the sample using the Private School Universe Survey. Appendix Table A2 shows that private schools in the sample tend to be larger and have higher student-teacher ratios than private schools not included in the sample.

[12]In Appendix Table A3 we show that our results are robust to different specifications of this regression. Specifically, we re-run our difference-in-differences where school value-added is estimated using Equation (1) without any demographic controls or prior test scores (Column 2), only including demographic characteristics (Column 3), including demographic characteristics and linearly controlling for prior test scores (Column 4).

[13]Private school test scores are normalized using only private and control public schools.

$testscore_{ist}$. In this specification, we cannot include third graders as they do not have a previous test score. γ_g and λ_g are grade-specific coefficients on lagged test scores and lagged test scores squared. $\mathbf{X}_i$ contains several indicators for student demographics including female, Black, Hispanic, Asian, two or more races, subsidized lunch, special education, Section 504, and testing accommodations. Our school value-added measure comes from the school-year fixed effects, β_{st}. The choice of the specification is motivated by that used in Chetty et al. (2014) to measure teacher value-added. Like Chetty et al. (2014), we control for grade-specific effects of lagged test scores to account for selection into particular schools. We also show in Appendix Table A1 that our results are robust to the use of an empirical Bayes shrinkage procedure in our value-added estimations (Kane and Staiger, 2008).

The primary concern surrounding the use of school value-added to proxy quality is that student sorting may bias our estimates if student's use of the voucher is correlated with some individual shock that artificially increases (decreases) school value-added in high exposure public (private choice) schools. We thoroughly discuss student sorting in Sections V.A and VI.A. Moreover, to mitigate these concerns for our public school sample, school value-added measures are estimated using students that never use a voucher and would not qualify for a 90% voucher. This sample selection does not fully address concerns over student sorting since we cannot identify public school students that qualify for a 50% voucher but significantly improves upon estimates that use the entire sample.

School value-added measures have emerged as a popular proxy for school quality in the literature on school choice programs (Betts and Tang, 2008; Allende et al., 2019; Neilson, 2021). The advantage to school value-added is that it isolates the incremental contribution of schools to student learning by controlling for prior achievement and other background factors. This approach mitigates biases associated with raw test scores or expenditure-based metrics that often reflect socioeconomic factors rather than school quality. Furthermore, some work has shown that value-added measures are weakly related to important long-term outcomes (Beuermann et al., 2022).

Appendix Figure A1 depicts the density plots of our school value-added estimates for both the public and private choice schools in our sample. Panel A shows the different distributions in the years before the policy was implemented, while Panel B plots our estimates in the years after expansion. For each panel, we report the p-value for the Kolmogorov-Smirnov equality-of-distributions test. In the years before the policy, the distribution of quality for private choice schools is the right of public schools. The p-value from the Kolmogorov-Smirnov test confirms that the two distributions are not identical. After expansion, the overlap of the two distributions is much greater. The following sections of this paper will separately analyze the changes in public and private choice school quality.

III.B Construction of Exposure Measure

Our main measure for each school's exposure to the voucher policy relies on the radial distance between the physical address of each of the public schools in the sample and all of the eventual private choice schools in Indiana open in the academic year prior to the adoption of the policy. A public school is considered to face high exposure to the voucher policy if the nearest eventual private choice school is within five miles of its location.[14] We find that around 60% of the public schools in the sample have at least one nearby eventual private choice school.[15] We show the full distribution of the distance between each public school in our sample and their nearest eventual private choice school in Appendix Figure A5. Public schools whose nearest eventual private choice competitor is outside the five-mile radius comprise our control group. Nearly all private choice schools (over 98%) are located within five miles of a public school; therefore, we distinguish between high exposure and control private choice schools by where they fall in the distribution of the number of public schools within five miles. High exposure private choice schools are those in the top tercile of this distribution, with the control group then making up the bottom two-thirds.[16]

Table 2 reports summary statistics for the high exposure and control schools in the academic year before the policy intervention. Columns (1)-(3) presents the sample means and differences for the public school sample, while Columns (4)-(6) report the same for the private choice school sample. Across both samples, high exposure schools are different from those in the control group on several dimensions. High exposure public schools were larger, with an average of 266 students taking the state exam versus 214 in control schools. They also had a smaller share of their students identified as White, 67% versus 92%; had a larger share of students identified as Black; 15% versus less than 1%; and had a larger share of students qualify for subsidized lunches, 54% versus 42%.[17]

[14] The public school results are robust to this definition of high exposure as shown in Panel A of Appendix Figure A2. We further bolster our definition oh high exposure by leveraging address data from voucher students. Using a random sample of 2,500 students, Appendix Figure A15 presents a histogram of the distance voucher students travel to attend a private choice school. We find that around 70% of voucher students travel less than 5 miles, supporting our definition.

[15] One might be interested in understanding whether our five mile, high exposure definition is supported by changes in enrollment. While a decline in enrollment places pressure on public schools, we are interested in the threat of competition rather than realized changes in the number of students served. Furthermore, enrollment is an equilibrium outcome as families likely choose schools based on their quality. We report that high exposure public schools witness increases in quality; therefore, we may expect to see no changes in enrollment following the adoption of ICSP. Nevertheless, we report the enrollment effects of the voucher program in Appendix Table A12.

[16] We show robustness to the definition of high exposure and the exclusion of baseline covariates for private choice schools in Appendix Table A10.

[17] The differences in the demographic make-up of the two groups of schools are at least partly explained by their locations within the state. Appendix Figure A6 shows the location of each public school in the sample. Public schools

The differences in demographic characteristics did not translate to significant differences in our outcome variables of interest in the public schools sample. High exposure public schools had an average overall school value-added estimate of -0.010 in the 2010-2011 academic year versus an average of -0.017 for the schools in the control group. In that same year, high exposure schools had an average school math value-added estimate of -0.005 and an average school reading value-added estimate of -0.023. Schools in the control group had an average of -0.006 and -0.045 in their school math and reading VA estimates, respectively. We find a similar pattern in the comparison between high exposure and control private choice schools. Importantly, our empirical strategy does not rely on the equality of the pre-policy summary statistics. Instead, identification requires that the change in outcomes for the control group are what those facing high exposure would have experienced had the policy not been put in place. We discuss this assumption in further detail in later sections.

IV Reduced-Form Empirical Strategy

To estimate the effects of introducing (and expanding) private school vouchers in Indiana we use a difference-in-differences model that relies on plausibly exogenous variation in a school's exposure to the voucher policy. We compare the change in school value-added in the years before and after the implementation of the policy in schools facing high exposure to the policy versus those in the control group. The underlying assumption in this strategy requires that, in expectation, the change in outcomes for the schools in the control group reflect what the schools facing high exposure would have experienced had the voucher policy not been implemented. While this assumption is ultimately untestable, we address this concern by reporting the results of an event-study specification that allows the effect of the voucher program to vary by years since implementation.

We implement this difference-in-differences (DID) strategy using the following regression:

$$VA_{st} = \beta_1 Post_t \cdot HighExposure_s + \sum_{t=2007}^{2018} \Psi_t(\mathbb{1}\{year = t\} * X_s^{2007}) + \delta_s + \gamma_t + \epsilon_{st} \qquad (2)$$

where VA_{st} is our constructed measure of value-added in school s at year t; $Post_t$ is an indicator that equals one in the years after the voucher policy was introduced; $HighExposure_s$ is an indicator that equals one if the public school is identified as having a nearby choice school; α_s is a school fixed effect that removes any time-invariant characteristics about the school that could otherwise bias our results; γ_t is a standard year fixed effect and ϵ_{st} is our idiosyncratic error term. Ψ_t captures the potentially time-varying effects of X_s^{2007}, a vector of initial school-level characteristics.[18] The

with a nearby private choice competitor are often located in the most populous and urban counties in Indiana, while those in the control group are spread out across the more rural parts of the state.

[18] In Appendix Table A4, we show our public results are in the same direction for overall and math VA when

parameter β_1 is the coefficient of interest and captures the average difference between the high exposure and control schools in the years after adoption of the voucher policy relative to the years before. All standard errors allow for arbitrary correlation in errors at the school level.[19]

We visually test the validity of the common trends assumption by presenting a set of event-study results that allow the effect of adopting a voucher policy to vary by years since implementation. Specifically, we run the following regression:

$$\begin{aligned} Y_{st} = \sum_{l=-5, l\neq -1}^{6} \theta_l HighExp_s \cdot \mathbb{1}\{t - 2012 = l\} + \sum_{t=2007}^{2018} \eta_t(\mathbb{1}\{year = t\} * X_s^{2007}) \\ + \pi_s + \lambda_t + \mu_{st} \end{aligned} \tag{3}$$

where l represents the lag or lead of interest, and 2012 is the year of adoption. Since we omit the year before the adoption of the policy, each θ_l captures the effect of being a school facing high exposure relative to the year before the introduction of the voucher program.

Our estimation strategy bypasses the concerns present in the current difference-in-differences literature because (1) we do not exploit variation across groups treated at different times (Goodman-Bacon, 2021); (2) our main specification relies on a binary measure of treatment (Callaway et al., 2024); and (3) we do not use time-varying covariates in any of our analyses (Caetano et al., 2022). Furthermore, adding school-level, time-varying characteristics may be inappropriate in this context. Characteristics such as the share of students eligible for subsidized lunches may change in the post-period as a direct result of the policy; hence their inclusion in our models would bias our results.

A potential weakness of our empirical approach is the possibility that private schools select whether to participate in ICSP based on differential trends across treated and control public schools. For example, if private schools in areas where average public school quality is falling select into the program, then our estimated competitive effects of ICSP would be downward biased by preexisting public school trends. After presenting our main results, we conduct an event-study analysis (as discussed above) to demonstrate that our results are not driven by differences in trends across treated and control public schools. We discuss private school selection into ICSP in greater detail in Appendix Section E1.

excluding baseline covariates are robust to the use of a continuous measure of the number of nearby eventual private choice schools.

[19]One may be concerned that our standard errors are incorrect in this specification as we are using an estimated variable as our outcome variable of interest. To address this issue, we perform a bootstrapping procedure as described in Appendix Section C1. We find that our estimates are more precise under this procedure, most likely because clustering at the school level significantly increases our standard errors. We, therefore, continue with our preferred specification.

V Effects of ICSP on Public School Quality

We begin by describing the estimated competitive effects of the Indiana voucher program on public schools with a nearby private choice school competitor. Appendix Figure A3 depicts the density plots of our school VA estimates for the public schools in our sample across two periods, pre-2011, and post-2013 to align with the policy time horizon. Panel A shows the kernel density plots for the high exposure public schools, and Panel B plots the data for the public schools in our control group. While not a formal difference-in-difference design, Appendix Figure A3 provides a visual preview of our findings.[20]

The results of our main analysis are reported in Table 3. Each cell represents the coefficient on the $Post_t \cdot HighExp_s$ interaction for separate regressions. Public schools with a private choice competitor within five miles saw an overall increase in their school value-added by 0.020 of a standard deviation in the post-policy period.[21] This increase in overall school value-added is driven by improvements in math value-added, with the estimate in Column (2) suggesting high exposure public schools witnessed a 0.028 s.d increase following the adoption of ICSP. High exposure public schools also saw a statistically insignificant increase in reading value-added (as shown in Column (3)); however, the event-study results do not show a clear improvement.

The result that the competitive effect of the voucher program induced an increase in school quality experienced by public school students is significant. Increased schooling quality is associated with better educational outcomes including increases in the likelihood of college attainment (Deming et al., 2014) and increases in the likelihood of attending a college with a larger share of STEM degrees (Shi, 2020). Therefore, our results not only suggest that voucher programs can induce responses by schools, but they can do so in such a way that meaningfully changes the educational outcomes of students not participating in the program.

Our findings on school quality also complement the results found in Waddington and Berends (2018) that explore the effect of ICSP on the students that use the voucher. The authors use a matched difference-in-differences design to compare students that used a voucher to those that qualified and remained in public schools. They find that voucher students see significant declines in math scores and no changes in reading scores following the switch to a choice school. While the authors do not speculate on the mechanisms that could explain their results, our estimates suggest

[20] While the p-value for the Kolmogorov-Smirnov equality-of-distributions test suggests the two distributions are not identical for the control group, we find no changes in school quality when running the difference-in-differences specification on the set of control schools (arbitrarily identifying high exposure as a choice school within 8 miles of its location). The results are shown in Panel B of Appendix Table A4.

[21] Results are are robust when we eliminate public schools that have a private choice school competitor within 3-8 miles of their location (Panel C of Appendix Table A4).

that the improvements in public school quality, particularly in math, can at least partially explain the declines they report.

ICSP was adopted and expanded in two separate academic years (2011-2012 and 2013-2014, respectively). One might then wonder if the two events had differential impacts on public-school quality. We answer this question using our event-study specification. The results of Equation (3) allow us to look at the effect (relative to the year before adoption) of facing private choice school competition in each year of the sample rather than averaging across the entire post-policy period. We can then compare the results at the year of expansion to that of the year of adoption to get a sense of which event is driving the results. Figure 1 shows a small jump in school math VA in the year of adoption of ICSP; however, the effects are largest in the years after ICSP's expansion. Interestingly, these results suggest that despite facing the threat of losing students as the program is adopted, public schools do not seem to respond until a much larger percentage of the student body qualifies to participate. This finding suggests that we may not expect voucher programs to have these indirect effects on educational outcomes until eligibility is broad.

While these estimates are modest in magnitude, they are statistically significant and indicate a positive relationship between the threat of private choice school competition and public school quality. Additionally, our results are likely a lower bound on the competitive effect of ICSP on public schools since even the control schools face some increased competition. We cannot make exact comparisons between our results and that of the extant literature as we are analyzing school VA rather than pure student test scores; however, our results are similar to the aggregated school-by-year estimates shown in Figlio and Hart (2014).

We have also estimated models at the student-school-year level and continue to see positive and statistically significant results on the effect of the threat of choice school competition on public school performance. These models are presented in Appendix Table A5 and show that our results are similar in size to those found in the first few years after the Florida voucher program was adopted (Figlio et al., 2023). Interestingly, our finding that the competitive effects of ICSP was stronger for math value-added versus reading value-added runs counter to studies examining the program in Florida, illustrating the importance of studying the competitive effects of voucher programs in different contexts (Figlio et al., 2023).

Our estimated improvements in public school value-added are modest to the gains in test scores documented in the literature examining the impacts of charter school expansions. For example, recent work by Ridley and Terrier (2023) and Gilraine et al. (2021) show that students exposed to charter school entry and expansion experience improvements in test scores around 0.02 to 0.05 standard deviations (see Cohodes and Parham (2021) for a review). We believe that at least some of the difference in magnitudes may be attributed to our comparisons of school value-added rather than test scores.

We also examine the impact of ICSP on students' non-academic outcomes. Columns (4)-(6) of Table 3 reports the results of Equation (3) where the outcomes of interest are school-level measures of attendance and disciplinary infractions. These two measures have been cited as important indicators for changes in behavior (Imberman, 2011a). After the implementation of ICSP, public schools facing the threat of private choice school competition saw increases in attendance with no changes in suspensions/expulsions. Specifically, high exposure public schools saw increases in attendance of 0.333 percentage points (p.p.), or about half a day, compared to those in the control group. The estimate in column (6) suggests that high exposure public schools saw no change in expulsions and suspensions. Attendance is cited as important determinant of student outcomes including test scores (Goodman, 2014; Fitzpatrick et al., 2011; Gottfried, 2009) and high school graduation (Liu et al., 2021). Using the estimates in Goodman (2014), we can do a back-of-the-envelope calculation that reveals that the increase in attendance by half a day, induced by ICSP, can translate into around a 0.025 s.d. deviation increase in test scores.[22] Overall, we take these results as evidence that in response to increased competition, public schools are increasing quality such that we see improvements beyond changes in test scores.

V.A Threats to Validity

The previous section shows that ICSP implementation is associated with increased school value-added estimates for public schools with a nearby private choice school. There remain, however, several potential threats to validity that should be addressed. Specifically, (1) student sorting may drive the improvements we find rather than actual changes made by the public schools, (2) the impact of the voucher policy on high exposure public schools may be driven by differential trends in school value-added across the high exposure and control groups before ICSP, (3) the results may be sensitive to the exclusion of particular districts that house a large proportion of the students in the state, and (4) there are other policy innovations beyond ICSP that may be driving the results.

We first investigate the issue of student sorting by documenting any changes in the demographic composition of students in high exposure public schools after the implementation of the program. Panel A of Table 4 reports the results of Equation (2) where the outcomes of interest are school-level measures of demographic variables (Share Female, Share White, Share Black, etc.) for public schools. After the implementation of ICSP, public schools facing the threat of private choice school

[22] One does need to keep in mind that the estimate from Goodman (2014) has a very specific interpretation, as it is identified off of missed classes due to snowfall, that may not translate to our context.

competition saw statistically insignificant changes of -0.29 p.p in the share of students that identify as female, -0.01 p.p in the share of students that identify as Black, and -0.1 p.p in the share of students qualifying for subsidized lunch when compared to the control group. However, as shown in columns (2) and (4), high exposure public schools saw a statistically significant decrease of 2.90 p.p in the share of White students and an increase of 2.61 p.p in the share of Hispanic students.[23]

Overall, we take these results as evidence that the demographics of students are changing with the implementation of the voucher program. To understand to what extent these changes in demographics drive our results, we perform an exercise with predicted school value-added outlined in Appendix Section D1. We find no evidence that high exposure public schools were predicted to improve their quality based on the change in composition of their students. In fact, we find that based on changes in observable characteristics, high exposure public schools were predicted to see declines in all of our outcome measures of interest. We take this result as strong evidence that it is changes made by schools that drive the improvements in quality we see. We also recognize that this exercise can only speak to how changes in observable school characteristics may have affected our school quality results. The concern remains that non-random student sorting on unobservable characteristics drive our findings.

We can mitigate some concerns of non-random sorting on unobservable characteristics by highlighting the strength of our value-added estimation strategy. In Equation 1, we control for lagged test-scores. Assuming that prior test scores fully proxy for those inputs that affect a student's achievement prior to using the voucher and that those inputs correlated with a student's likelihood of using a voucher, we address the concerns for this type of sorting. This is a strong assumption; however, it is standard in the school value-added literature.

The concern over changes in student composition stems from the idea that these changes impact our estimates of school value-added through peer effects. In an attempt to shut down the peer effects channel, we use student-class links available in the IDOE-CREO database to create a sample of students in public schools that did not attend classes with any ever voucher students and did not qualify for a 90% voucher themselves. The school value-added estimates stemming from this group of students would ameliorate this concern if we assume that for a given student, peer effects are driven solely by the other students in their classes. A caveat to this exercise is that information on

[23]Often of interest is the question of whether ICSP induces "cream-skimming", the movement of the best public school students to the private sector. Appendix Figure A4 shows the density plots of standardized test scores in years before the program was adopted for students who eventually use a voucher and those students who remain in the public-school system despite qualifying to participate. We find that eventual voucher students slightly outperformed those remaining at the public schools, suggesting a slight positive selection of voucher students. However, the extent of the sorting depends on the exact comparisons made as shown in Appendix Figure A8.

student-class links is only available from the 2010-2011 through 2017-2018 academic years, which limits the sample to only one year of pre-policy data. The classroom data in the 2010-2011 year is also limited due to inconsistencies in data collection. Nevertheless, Panel F of Appendix Table A4 reports the results of our difference-in-difference strategy using school-value added estimates from this sample. We continue to find that high exposure public schools saw improvements in quality, suggesting that changes in peer effects are unlikely to be driving our results.

To ensure that the findings are not driven by differential trends between the schools facing high exposure to the voucher policy and the control group, Figure 1 plots the event-study results. Prior to implementation, high exposure public schools and the control group appear to have similar trends in school value-added, shown by the relatively flat differences between the two groups.[24] In all years before implementation, the 95 percent confidence interval contains zero, which means that in those years, the difference between high exposure and control groups cannot be distinguished from the value in the year before implementation. Furthermore, Appendix Figure A7 reports the results of the event-study sensitivity analysis as proposed by Rambachan and Roth (2023). Our breakdown value for a significant effect on overall and math school value-added in the year the ICSP program expanded (2013-2014 academic year) is equal to one, meaning our result is robust to allowing for violations of parallel trends up to the max violation in the pre-treatment period.

The third concern is that the results are sensitive to the exclusion of particular school districts. We, therefore, estimate the main analysis in Table 3 excluding Marion County, the largest county in the state and the home of Indianapolis. We find consistent evidence that, regardless of dropping Marion County, the signs and general significance levels of the interaction term of interest hold as shown in Panel E of Appendix Table A4. Appendix Figure A14 shows that when we drop any of the 92 Indiana counties, our results remain similar to the full-state analysis. Therefore, it is difficult to believe that some combination of specific counties are driving the general direction of our results.

Another concern is that other policy interventions beyond the voucher program are driving the results. To address this issue we use year fixed effects in each of our specifications to capture shocks common to both the treatment and control groups. Unaccounted for shocks could still exist, but those shocks would have had to elicit disproportionate reactions from schools with a nearby choice competitor to account for our results. A particular concern is that in 2011 the implementation of the Teacher Evaluations and Licensing Act and the introduction of Indiana's A-F school grading system may have affected school quality. However, since the quality of schools in the high exposure and

[24] We further show the robustness of our results using placebo treatment years. Appendix Figure A9 shows the results when we assign the adoption of ICSP to be two years prior to the actual The figure shows that school quality only improved following the years of actual adoption and expansions (As indicated by the red and blue dashed lines, respectively).

control groups were statistically indistinguishable in 2010, it is unlikely that either of these reforms differentially impacted the two sets of schools. Moreover, it is not clear whether schools felt increased pressure to improve quality as a result of these accountability programs. Prior to the adoption of these specific measures, schools and teachers were held to other accountability metrics. Furthermore, in the 2013-2014 academic year, less than 0.5 percent of teachers were cited as "ineffective" and only 4 percent of public elementary and middle schools were given an "F" grade (Indiana Department of Education, 2014b,a).

V.B Public School Heterogeneity by School Attributes

We have found consistent evidence of modest improvements in school VA when comparing public schools facing the threat of private choice school competition to those in the control group. However, these average estimates across all public schools facing competition could differ across various subgroups. Therefore, we disaggregated the results by the following baseline characteristics: free/reduced price lunch enrollment, total enrollment, overall school VA, median income of the census block group where the school is located, and overall school VA of nearest eventual private choice school. We calculated these estimates by introducing interactions of the school subgroup with the $Post_t \cdot HighExp_s$ indicator in Equation (2).[25]

Table 5 displays the results of our heterogeneity analysis by school subgroup for overall school VA, school math VA, and school reading VA, respectively. Panel A show the differences in outcomes for public schools with above- or below- median enrollment of students who qualified for subsidized lunch in the year before ICSP was introduced. The interaction term in Panel A isolates the competitive impact on the set of schools facing the highest threat of competition, since these public schools are located within five miles of a private choice school and enrolled a large share of students that would automatically qualify for the voucher. We find that across all of the outcome variables of interest, improvements in school quality are driven by the set of high exposure schools that felt additional competitive pressure because a large share of their enrollment automatically qualified for the voucher. These results bolster the claim that changes in school VA following the adoption of ICSP are due to increased competition.

Panel B displays the differences in outcomes for public schools with above- and below-median enrollment for the 2006-2007 academic year. Across all of the columns, the estimate on the interaction

[25] We have also considered heterogeneity by initial levels of suspension/expulsions. This analysis addresses a different type of threat public schools could face. Specifically, families may have a desire to leave public schools that they deem unsafe. We do not find any differential effects for public schools that had an above median percentage of their students ever being suspended or expelled. Results are available upon request.

term with above-median baseline enrollment is statistically insignificant. This result implies that public schools see similar improvements in quality when facing the potential threat of competition regardless of whether they have relatively small or large baseline enrollment.

In Panel C, we examine the differences in outcomes for public schools with above- or below-median overall school value-added for the 2006-2007 AY. Across all outcome variables of interest, the estimate on the interaction term with above-median baseline school VA is negative, statistically significant, and almost equal in magnitude to the overall estimate on the $Post_t \cdot HighExp_s$ indicator. These findings imply that the changes we see in school quality are driven by the schools that face potential competition and were originally low-performing. In fact, high exposure schools with above-median baseline school value-added see small or no changes in the outcomes of interest when compared to the control group. The increase in school quality for low-performing schools, coupled with the null results for high-performing schools, suggests that the gap in public-school quality is closing as a result of the program.[26]

Panel D reports the effects on quality by the mean income of the census block group where the public school is located. This specification allows us to capture any differences in the results between public schools located in relatively rich versus poor neighborhoods. Similar to the results in Panel B, the estimate on the interaction term with above-median neighborhood income is statistically insignificant across all outcomes of interest. These findings imply that schools see similar improvements in quality when facing the potential threat of competition regardless of whether they are located in a relatively poor or rich neighborhoods.

Panel E presents the differential impacts on quality by the overall school VA of the nearest private choice school. Specifically, the interaction term includes an indicator for whether the nearest competitor had a baseline overall school VA that was greater than that of the public school. Under the assumptions that (1) families value higher quality schools, as evidenced by the structural literature (Hastings et al., 2009; Allende, 2019; Neilson, 2021), (2) the voucher program eliminates price differences between the public and private choice school (45% of voucher students paid zero dollars in out of pocket expenses during our sample period) and (3) the distance traveled to attend school remains relatively fixed, these public schools would face a greater competitive threat. We find, across all of our outcome variables of interest, that public schools with a nearby, higher quality private choice competitor indeed witness the largest increases. We take these results as strong evidence that the increases in school quality we document for public schools is driven by the competitive threat

[26] One may be concerned that these results are driven by families wishing to leave low performing public schools; however, as shown in Appendix Figures A10 there does not seem to be differential student sorting on ability across these two types of public schools when comparing either FRPL or non-FRPL students.

these schools face from the voucher program.

We also explore possible heterogeneity by financial incentive. As shown in Figlio and Hart (2014), not all public schools face the same incentives to respond to the implementation of a voucher program. Specifically, public schools on the margin of receiving federal Title I aid may experience a larger reduction in resources as a consequence of losing students to private schools. We, therefore, explore whether high exposure public schools with Title I funding drive our results. Panel A of Appendix Table A6 reports the differences in outcomes for public schools with and without a Title I program in year before ICSP was adopted. Panel B of Appendix Table A6 includes an interaction term that allows us to identify the differential impact of ICSP on high exposure public schools that just qualified for Title I funding.[27] Overall, we do not find evidence that public schools facing greater financial pressure respond more to the program.

V.C Potential Mechanisms

V.C.1 Changes in School Inputs

Given the improvements we find in pubic-school quality, we next examine changes in schools inputs that might lead to increases in school quality. In particular, we combine information from the Common Core of Data on Indiana public schools from the National Center of Education Statistics with available teacher data in the IDOE-CREO database to explore changes in student-teacher ratios, the number of teachers with a high-quality (HQ) certification,[28] number of teachers with a graduate degree, teachers' average years of experience, number of teachers that identify as non-White and teacher retention rates. Unfortunately, the information on teachers is only available from the 2010-2011 through 2017-2018 academic years, which limits our sample to include only one year of pre-policy data.

Figure 2 separately plots the average of each of these school inputs across the available years of data for high exposure and control public schools. We do not find strong evidence that high exposure public schools saw meaningful changes in student-teacher ratios,the average years of experience of their teachers or the number of non-White teachers when compared to the control group. However, Panel B shows that while both high exposure and control public schools added around 2 additional HQ certified teachers (either through hiring or certification) in the year ICSP was adopted, control

[27] Title I funding is allocated based on where a school ranks within their district with respect to the share of low-income students they serve. In Indiana, schools that meet or exceed the district's poverty average are eligible to receiving funding. We define "just qualifying" for Title I as being within 5 percentage points above that cutoff for eligibility.

[28] High-Quality certification is determined by standards set by No Child Left Behind. States can add their own requirements. In Indiana, certification requires passing an exam to indicate proficiency in a certain subject.

public schools did not retain them. By the end of the sample period, control public schools had returned to their initial levels of HQ certified teachers. Panel C, shows that while both high exposure and control public schools see declines in the average number of teachers with a graduate degree, control public schools witness faster declines over the sample period. Furthermore, Panel F shows that while high exposure public schools had initially lower rates of teacher retention, the gap closed significantly following the adoption of ICSP.

We confirm these patterns in the data with the results from our difference-in-differences specification. In Table A7, we report the results of Equation 2 using school inputs as our outcome measures of interest. We find that relative to the year before ICSP was adopted, high exposure public schools saw increases of around 0.6 teachers with a graduate degree, 1.41 teachers with a HQ certification and an increase of 2.1 p.p. in the teacher retention rate when compared to the control group.[29] These changes in average teacher characteristics are significant. While the previous literature on the effects of advanced degrees on student outcomes is mixed, recent work shows that subject-specific teacher credentials (such as a high-quality certification) are associated with stronger student achievement (Strøm and Falch, 2020). Furthermore, high teacher turnover rates have been shown to be detrimental to the quality of instruction in public schools, suggesting improvements on this front would likely increase school quality (Hanushek et al., 2016).

We also consider the possibility that our results are driven by students working harder in response to the voucher program to gain admission into participating private schools. While we do not have the data to test this hypothesis specifically, there are several reasons we do not believe this mechanism fully explains our results. First, after reviewing admissions policies from several private choice schools, we found that the ranking of students is often determined by their relationship to the church that runs the school. Second, our discussions with school administrators revealed that private choice schools did not often face the space constraints that would incentivize students to perform better on the standardized tests in order to gain admission. Third, our school value-added measures are estimated off students that never participated in the program and would not qualify for a 90% voucher. While this sample restriction does not rule out that students qualifying for a 50% voucher could be working harder to gain admission, we see limited support for our results being driven by changes in student work effort.

[29] Our results differ from those in Figlio and Hart (2014). The authors find that schools faced with greater competition shift their teacher workforce to include less- qualified teachers. Unfortunately, we lack the detailed data on school practices to fully disentangle different school responses under each of these reforms.

V.C.2 Changes in School Financial Resources

ICSP could further have a direct effect on public schools' ability to improve school quality through changes in financial resources. Opponents of school choice policies argue that these programs drain public school finances through direct cuts in state funding (Strauss, 2017). Moreover, losing students eligible for subsidized lunches could result in further resource reductions if schools rely on Title I funding. By contrast, per-pupil revenue may increase in public schools if total federal and local funding remain unchanged. If the latter is the case in Indiana, increases in available school funds could contribute to our results.[30]

However, school funding in Indiana heavily relies on state rather than local sources. The state currently ranks 40$^{\text{th}}$ in the percent of public school funding coming from local revenues (just below 30%) (U.S Census Bureau, 2021). Furthermore, the state has provided 100 percent of funds available to support education-related operating costs since 2009. Local funds are used to support other expenses including transportation, capital projects, and debt services (Chu, 2019). This reliance on state-funding suggests that Indiana public schools are susceptible to reductions in revenues as students use the voucher. Anecdotal evidence from statements made by public school boards echo this concern (Gore et al., 2011). Unfortunately, school-level finance data is not available for a majority of our sample period; therefore, we cannot formally test whether changes school funding can explain our results.

Prior work examining the fiscal impacts of school choice policies have shown that these programs largely do not affect public school finances. For example, DeAngelis and Trivitt (2016) shows that if the Louisiana Scholarship Program was eliminated only 2-7 out of 69 school districts would see an increase in financial resources. In the charter school context, Ridley and Terrier (2023) shows that the expansion of these schools in Massachusetts left districts' overall per-pupil revenue and expenditure unchanged with shifts in spending from capital investment and support services to instruction and salaries. The shift in school spending led to increases in non-charter students' achievement in math, suggesting a plausible mechanism for our results.

VI Effects of ICSP on Private Choice School Quality

Our results thus far have been centered on public schools' responses to the competitive effects of ICSP. We next assess whether participating private schools also saw changes in school quality as a

[30] There still remains some debate on whether increases in school spending improve educational outcomes (Jackson, 2020). One direct way increased school per-pupil expenditure could directly influence our results is if schools used the extra funds to hire or convert high quality teachers.

result of the program. This investigation is necessarily more speculative than our analysis of public schools due to data constraints. We are unable to compare changes in outcomes for participating and non-participating private schools, since non-participating private schools often do not administer the ISTEP+ exam. We are also unable to leverage variation in when a private school starts accepting voucher students as a large percentage adopt in the first year of the program. However, in this section we present suggestive evidence that private choice schools are reducing quality in response to the competitive effects of ICSP.

We first investigate private choice schools' response to the adoption of the voucher program by plotting the averages of our school value-added measures for each year in the sample. Figure 3 plots these averages for our measures of school quality from 2007 through 2018. In the first year of the program, there is an immediate drop in average quality on all dimensions. This drop is most apparent for math value-added, but by the following year, the average reading value-added for private choice schools saw a similar decline. These school quality measures, while steadily increasing after 2013, remain below the pre-period levels throughout the sample period. While we do not assert any causal claims from this figure, it does suggest that private choice schools saw a decline in quality following the implementation of the program.

Because the adoption of ICSP reduces the price to attend a private choice school, these schools face a larger market of potential enrollees. Private choice schools surrounded by a larger number of public schools see the largest increase in potential students. Therefore, to assess the competitive effect of ICSP on private choice schools we compare changes in quality for private choice schools with a large pool of public schools to pull enrollments from to those with fewer public schools in the area.[31] High exposure is now defined as being in the top tercile of the distribution of the number of public schools within a five-mile radius. Appendix Figure A3 shows the density plots of our school VA estimates for these groups of private choice schools across two time periods: Pre-2011 and Post-2013 to align with the program's adoption and expansion. Panel A shows the kernel density plots for high exposure choice schools and Panel B plots the data for those choice schools in the control group. Both groups witness a leftward shift in the distribution of overall school value-added following the expansion of ICSP, suggesting that ICSP may not have elicited differential responses across our measure of exposure.

Table 6 formalizes this comparison using our difference-in-differences specification (similar to

[31] Appendix Table A10 alters the definition of high exposure for private choice schools. Rather than distinguishing treatment and control based on the distribution of the number of public schools within five miles, (1) we split private choice schools by the percentage of public school students that would qualify for the voucher in the schools within five miles of their location and (2) Define high exposure by whether the private choice school is within five miles of an initially low value-added public school. We find similar results under these specifications.

Equation (3)). Each cell in the first row of the table represents the coefficient on the $Post_t \cdot HighExp_s$ interaction for separate regressions. In the second row, we include an interaction term to indicate whether a private school with a large number of public schools within five miles also was of higher quality than the nearest public school in the baseline year. Columns (1) and (2) present the results on overall school VA; columns (3) and (4) present the results on school math VA; and columns (5) and (6) present the results on school reading VA. After the adoption of ICSP, high exposure private choice schools saw statistically insignificant decreases of around 0.02 s.d. across each of our measures of school quality when compared to the control group. The even columns show that this result is driven by those private choice schools with a large number of public schools within five miles and had higher baseline quality than the nearest public schools albeit with results remaining statistically insignificant. These findings serve as the reverse of those shown in our discussion of public school heterogeneity results. This exercise ultimately cannot explain the large drops in school quality seen in Figure 3, but suggests that private choice schools with a larger pool of students to pull from saw larger drops in school quality.

We also present the results of our event-study analysis for private choice schools in Appendix Figure A12. While we see declines in our outcome measures of interest in some years after the adoption of ICSP, the estimates are generally noisy. While we fail to identify a clear change in school value-added across our high exposure and control private choice schools, our descriptive statistics and difference-in-differences results are consistent in showing a small decline in school quality. Together, we take our results as suggestive evidence that the addition of potential enrollees into private choice schools due to the adoption of ICSP lead to small declines in our outcome measures of interest.

VI.A Threats to Validity

Similar to our public school results, we want to address potential threats to validity to our private choice school findings. First, one may be concerned that the declines in school quality we document are due to a negative selection of which schools participate in the program. It is difficult to directly compare school value-added across schools in and out of the sample, since a vast majority of private schools out the sample do not have test score data. However, for private schools that are accredited (which includes all private choice schools) we can examine differences in the grades schools receive by the Indiana Department of Education (IDOE) prior to the voucher program. In the 2010-2011 academic year, an overwhelming majority of private choice schools in our sample received an "A" grade (150 out of 179), which was higher than the corresponding share in private schools not included in the sample (60 out of 86).[32] This results suggests there is limited negative selection into becoming a private choice school in our sample.

In addition to these descriptive statistics, Table 7 includes additional heterogeneity analyses that sheds light on whether lower quality private choice schools drive the declines in school quality we find. Specifically, Table 7 disaggregates the main private choice school results by several pre-policy characteristics indicative of quality: declines in enrollment from 2007 to 2011, tuition costs, receiving a non-"A" grade in 2007 and baseline school value-added.[33] While we find do not find strong evidence of heterogeneous effects across changes in enrollment or levels of tuition, high exposure private choice schools that were initially low-performing (either given a non-"A" grade of had below median baseline school VA) saw marginally, statistically significant improvements in our outcome measures of interest. These results suggest that the declines we report in private choice school quality are driven by initially high performing schools, further assuaging concerns of negative selection.

One may also be concerned that the declines in private choice school quality are driven solely by changes in student composition. Indeed, Appendix Figure A11 shows that students who eventually use a voucher performed worse on standardized tests compared to students already attending the private choice schools. However, if changes in student composition drive the declines in school quality we document, we would then expect in years of improving quality (2014-2016) to see large exits of voucher students from private choice schools back to the public school system. Appendix Figure A13 shows that this is not the case. Specifically, we find that the years of improving private choice school quality correspond to the years of lowest rates of returning to the public school system. We take these findings as suggestive evidence that the declines in quality are not solely due to changes in student composition.

To fully address the concerns of student sorting, we begin by documenting changes in the demographic composition of students in high exposure private choice schools versus the control group. Panel B of Table 4 reports the results of Equation (2) where the outcomes of interest are school-level measures of demographic variables for private choice schools. After the implementation of ICSP, private choice schools facing with a relatively large number of public schools to pull students from saw a statistically insignificant changes of 0.59 p.p. in the share of students that identify as female, and a 1.38 p.p. in the share of students that identify as Black when compared to the control group. However, as shown in columns (2), (4) and (5), high exposure private choice schools saw a statistically significant decrease of -6.76 p.p. in the share of White students, an increase of 5.06 p.p. in

[32]The figure for out-of-sample private schools only includes schools with grades from IDOE. Around 70 additional private schools did not have a corresponding grade in the reference year.

[33]The analysis in Table 7 is similar in style to what is found in Abdulkadiroğlu et al. (2018) when exploring private schools selection as a mechanism behind their main findings.

the share of Hispanic students, and an increase of 4.46 p.p. in the share of students that qualify for subsidized lunch.

We take these results as evidence that the demographics of students are changing with the implementation of the voucher program. In Appendix Section D1, we repeat the predicted school value-added exercise for the private choice schools. We find that based on changes in observable characteristics, high exposure private choice schools were predicted to see statistically insignificant declines in overall value-added, increases in math value-added and no changes in reading value-added. These results suggest that part of the declines in school quality we report is due to changes in student composition. However, the results discussed in the following section highlight that changes in school inputs also likely play a role in explaining why there is a decline in private choice school quality following the adoption of ICSP.

We also examine the sensitivity of our results to the definition of high exposure for private choice schools. Appendix Table A9 shows that our results are qualitatively similar when we define high exposure as either being in the top quartile of the number of public schools within five miles (Panel A) or use a continuous measure of the number of public schools within five miles (Panel B). These results show that our findings do not hinge on the use of the tercile definition of high exposure and are robust across several specifications.

VI.B Mechanisms

To understand what is driving the declines in quality we find, we use data from the Private School Universe Survey to examine changes in private school inputs. Specifically, we have information on the number teachers, student-teacher ratios, and the time spent in school (in hours) every other year from 2006 until 2018. Figure 4 plots the averages of these inputs separately for high exposure and control private choice schools. We find that following the adoption of ICSP, there is evidence that high exposure private choice schools experienced an increase in their student-teacher ratios. Panel C shows that while both high exposure and control private choice schools increased their average in instructional time, control private choice schools saw a more significant rise. We confirm these findings with the results from our difference-in-differences specification shown in Table A8. We find evidence that following the adoption of ICSP, high exposure private choice schools saw a statistically significant increase of 0.83 in their student-teacher ratio (off a base mean of 14.22) compared to the control group with the results on the number of teachers and instructional time being statistically insignificant.

The results on student-teacher ratios is of similar magnitude to those found in Rinz (2015), which examines changes in private school inputs following the adoption of several voucher programs throughout the 2000s. The author's analysis includes both traditional voucher programs and large scale tax credit programs, which shows that these two variations of voucher programs may have similar impacts on private school responses. Evidence from Project STAR reveals that changes in student-teacher ratios can have a significant impact on student outcomes, including test scores (Krueger, 1999), high school graduation (Finn et al., 2005), college entrance exam taking (Krueger and Whitmore, 2001), college matriculation (Chetty et al., 2011), criminal activity, and teen birth rates (Schanzenbach, 2006). Therefore, our result that students in high exposure private choice schools experience increases in their class sizes further shows that the competitive effects of voucher programs can have important impacts on the educational outcomes of students that do not participate in the program.

VII Conclusion

This paper shows that the competitive effects of a large voucher program can lead to meaningful changes in school quality. We examined the effects of the adoption and expansion of the Indiana Choice Scholarship Program, one of the largest programs in the United States providing private school vouchers to low and middle-income families, and found that both public and private choice schools saw changes in their school value-added.

We found that public schools facing high exposure to ICSP experienced increases in their school quality, while private choice schools witnessed declines. Our estimates were modest in magnitude; however, papers evaluating voucher policies have found relatively small effects on student outcomes ranging from -0.01s.d. to 0.11s.d (Rouse and Barrow, 2009).[34] Furthermore, Figlio et al. (2023) shows that the impact on public schools grows as voucher programs mature. We analyze the program in the first few years of its adoption, so it is possible there will be stronger increases in the future.

Our results complement those found in previous work examining the effect of ICSP on students that use the voucher. Waddington and Berends (2018) shows that students participating in the program saw declines in math performance with no changes in reading. We argue that schools' responses can at least partly explain these student-level results. The results in Waddington and Berends (2018) might overstate the decline in math performance since this is the dimension that high exposure public schools saw the greatest improvements. Our results provide an example of how the understanding of a program's effectiveness may change when we take into consideration the indirect effects as a policy greatly expands eligibility.

Combining our results on public and participating private schools with previous work on the

[34] Abdulkadiroğlu et al. (2018) and Waddington and Berends (2018) are notable exceptions.

effects of ICSP on participating students, we can begin to get a better sense of the total impact of the voucher program in Indiana. While we abstain from making claims on changes in total welfare - as we do not want to make any assumptions about consumer's utility functions or school's objective functions - we do know that a much larger share of students attend public schools ($\sim$89% of the population aged 5-18 in 2018) than accredited private schools ($\sim$7% of the population aged 5-18 in 2018) in Indiana. This descriptive fact suggests that the improvements in public school quality may outweigh the declines in private school quality we find, but that conclusion depends on the goals of the state of Indiana and financing of the program. Recent work evaluating the welfare impacts of a voucher program in India suggest that there can be substantial welfare gains from the adoption of these policies when the cost of public schools are significantly higher than private schools and programs are targeted (Arcidiacono et al., 2024). Future work will examine whether similar improvements can be estimated in the context of Indiana.

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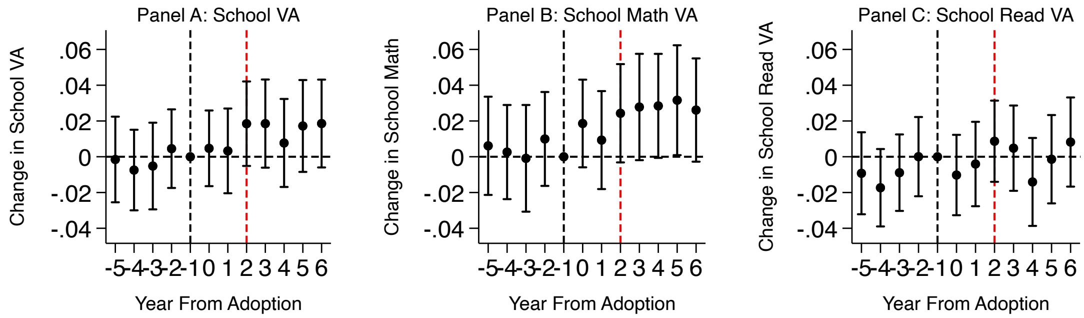
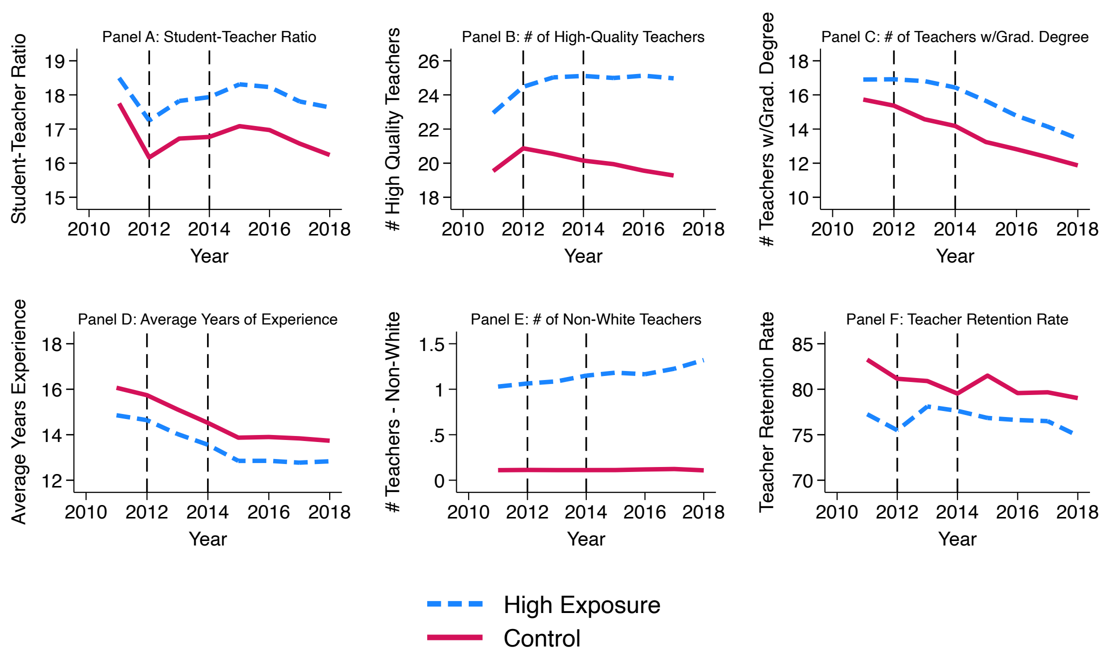
Figures

Figure 1: Public School Event-Study Results



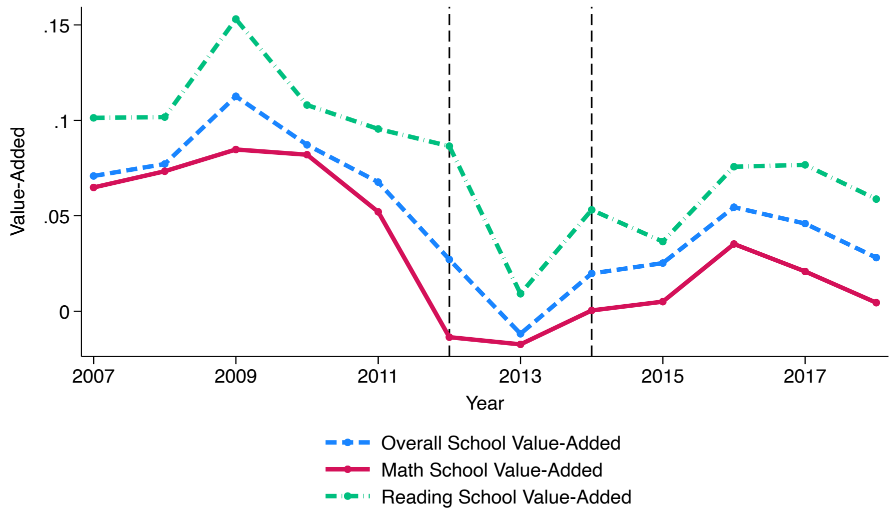
Notes: This figure presents the event-study estimates from Equation (3). Figure 1(a) plots the estimates for overall school value-added, Figure 1(b) plots the estimates for school math value-added and Figure 1(c) plots the estimates for school reading value-added. Each figure is the result of a separate estimation. 95% confidence intervals are reported. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. High exposure is defined as having at least one eventual private choice school within five miles. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).

Figure 2: Public School Inputs



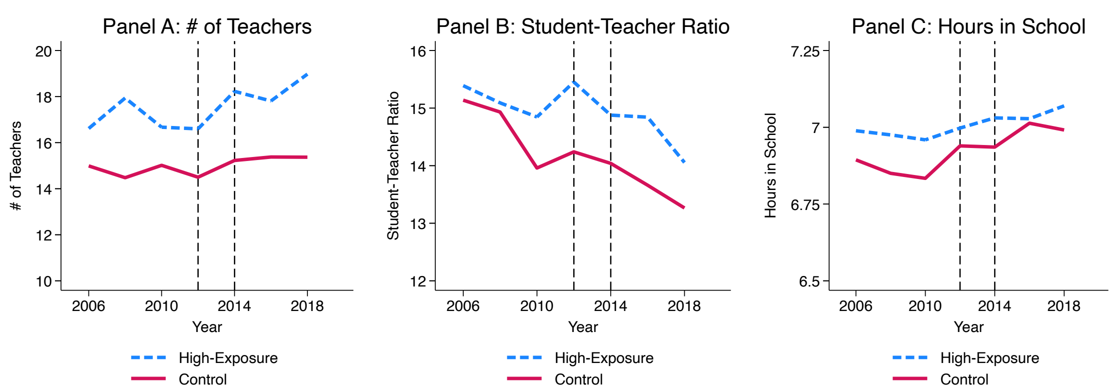
Notes: This figure presents the average student-teacher ratio (Panel A), number of high-quality teachers (Panel B), number of teachers with a graduate degree (Panel C), average years of experience of teachers (Panel D), number of non-White teachers (Panel E) and average teacher retention rate (Panel F) across public schools in the sample. High-exposure is defined as having an eventual private choice school within five miles of the public school's location. Data on student-teacher ratios come from the Common Core of Data from the National Center of Education Statistics. Data on teacher characteristics come from the IDOE-Database.

Figure 3: Private Choice School Value-Added



Notes: This figure depicts the average school value-added (VA) estimates across all private choice schools in each year of the sample. School VA estimates are calculated using the OLS regression described by Equation (1). Data on test scores and enrollment come from the IDOE-CREO database. The black, dashed lines represents the years the voucher program was implemented and expanded.

Figure 4: Private Choice School Inputs



Notes: This figure depicts the average number of teachers (Panel A), the average student-teacher ratio (Panel B), and the average hours spent in school (Panel C) across high-exposure and control private choice schools in the sample. High-exposure private choice schools are those in the top tercile of the distribution of number of public schools within five miles of the private choice school's location. Data on private choice school inputs come from the Private School Universe Survey conducted by the National Center for Education Statistics. Data are only available in every other year. The black, dashed lines represent the years the voucher program was implemented and expanded.

Tables

Table 1: Descriptive Statistics on the Indiana Choice Scholarship Program

Academic Year	Income Limit for Family of Four	Number of Participating Students in K-12	Fraction of K-12 Students Participating	Number of Participating Schools	Net Government Payment
2011-2012	62,022	3,911	0.40%	241	$15,514,024.60
2012-2013	63,965	9,139	0.80%	289	$36,042,923.03
2013-2014	87,135	19,809	1.80%	313	$78,593,339.80
2014-2015	88,245	29,148	2.60%	314	$112,707,313.24
2015-2016	89,725	32,686	2.89%	316	$131,514,681.60
2016-2017	89,910	34,299	3.03%	313	$142,193,570.25
2017-2018	91,020	35,458	3.11%	318	$151,377,354.18

Notes: This table presents descriptive statistics about the Indiana Choice Scholarship Program using data collected from the Indiana Choice Scholarship Reports published by the Indiana Department of Education. The income limits are reported in nominal dollars for students to receive a 50% voucher. Net government payments are also reported using nominal dollars.

Table 2: Summary Statistics of High Exposure vs. Control Schools

	Public Schools			Private Choice Schools		
	High Exposure	Control	Difference	High Exposure	Control	Difference
	(1)	(2)	(3)	(4)	(5)	(6)
# of Students Taking ISTEP+ Exam	266	214	-52***	147	107	40**
	(240)	(172)		(87)	(71)	
School VA	-0.010	-0.017	-.007	0.054	0.064	0.010
	(0.190)	(0.163)		(0.184)	(0.164)	
School Math VA	-0.005	-0.006	-0.001	0.034	0.054	0.020
	(0.213)	(0.195)		(0.228)	(0.179)	
School Reading VA	-0.023	-0.045	-.022*	0.084	0.096	0.009
	(0.191)	(0.147)		(0.130)	(0.174)	
% White	66.588	92.575	25.987***	74.393	88.800	14.407***
	(27.205)	(7.228))		(27.398)	(12.484)	
% Black	14.815	0.647	-14.168***	8.125	1.591	-6.534**
	(20.367)	(1.301)		(16.128)	(4.541)	
% FRPL	52.973	41.817	-11.156***	23.047	11.603	-11.444**
	(25.303)	(14.149)		(26.232)	(12.299)	
Number of Schools	785	484	1,269	53	126	179

Notes: This table presents summary statistics for the set of schools identified as either high exposure or control in the year before the voucher policy was implemented. For public schools, high exposure is defined as having at least one eventual private choice school within five miles. For private choice schools, high exposure is defined as being in the top tercile of the distribution of number of public schools within five miles. Columns (3) and (6) reports the difference in the means between schools in the control group and those facing high exposure to the policy. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).

Table 3: The Competitive Effects of ICSP on Public Schools

	(1) School Value-Added	(2) School Math Value-Added	(3) School Reading Value-Added	(4) Percent Days Attended	(5) Total Days Attended	(6) Percent Expelled or Suspended
$Post_t \cdot HighExp_s$	0.020***	0.028***	0.009	0.333***	0.543***	0.002
	(0.006)	(0.008)	(0.005)	(0.010)	(0.185)	(0.002)
Observations	15,228	15,228	15,228	15,216	15,216	15,216
R-squared	0.453	0.426	0.443	0.844	0.836	0.773

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. High exposure is defined as having at least one eventual private choice school within five miles. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).

Table 4: DiD Results on Demographics of Enrolled Students

VARIABLES	(1) Share Female	(2) Share White	(3) Share Black	(4) Share Hispanic	(5) Share FRPL
Panel A: Public Schools					
$Post_t \cdot HighExp_s$	-0.289	-2.908***	-0.008	2.610***	-0.100
	(0.189)	(0.244)	(0.161)	(0.213)	(0.322)
Observations	15,228	15,228	15,228	15,228	15,228
R-squared	0.295	0.977	0.971	0.927	0.946
Panel B: Private Choice Schools					
$Post_t \cdot HighExp_s$	0.593	-6.755***	1.382	5.061***	4.459**
	(0.872)	(1.748)	(0.948)	(1.697)	(2.112)
Observations	2,148	2,148	2,148	2,148	2,148
R-squared	0.265	0.909	0.918	0.893	0.834

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that receive testing accommodations in the 2006-2007 academic year. High exposure is defined as having at least one private eventual choice school within five miles. Data on enrollment come from the IDOE-CREO database.

Table 5: Public School Heterogeneity Results

VARIABLES	(1) School Value-Added	(2) School Math Value-Added	(3) School Reading Value-Added
Panel A: High Baseline FRPL Enrollment			
$Post_t \cdot HighExp_s$	0.009	0.014*	0.002
	(0.007)	(0.009)	(0.006)
Interaction with Above	0.026***	0.032***	0.016*
Median Baseline FRPL Enrollment	(0.007)	(0.011)	(0.008)
Observations	15,228	15,228	15,228
Panel B: Large Baseline Enrollment			
$Post_t \cdot HighExp_s$	0.006	0.012	-0.006
	(0.008)	(0.010)	(0.007)
Interaction with Above	-0.011	-0.009	-0.009
Median Baseline Enrollment	(0.008)	(0.009)	(0.007)
Observations	15,228	15,228	15,228
Panel C: High Baseline School Value-Added			
$Post_t \cdot HighExp_s$	0.045***	0.059***	0.027***
	(0.007)	(0.009)	(0.006)
Interaction with Above	-0.047***	-0.057***	-0.035***
Median Baseline School VA	(0.008)	(0.009)	(0.007)
Observations	15,228	15,228	15,228
Panel D: Above Median Neighborhood Income			
$Post_t \cdot HighExp_s$	0.025***	0.035***	0.012*
	(0.008)	(0.010)	(0.007)
Interaction with Above	-0.009	-0.012	-0.005
Median Neighborhood Income	(0.008)	(0.010)	(0.007)
Observations	15,228	15,228	15,228
Panel E: Nearest Private Choice School Has Higher Quality			
$Post_t \cdot HighExp_s$	-0.017	-0.014	-0.022**
	(0.011)	(0.013)	(0.009)
Interaction with Closest Competitor	0.055***	0.064***	0.045***
Has Higher Baseline Quality	(0.010)	(0.012)	(0.009)
Observations	10,416	10,416	10,416

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. High exposure is defined as having at least one eventual private choice school within five miles. A high (small) baseline FRPL school is defined as one that has above (below) median FRPL enrollment in the 2010-2011 AY. A small (big) school is defined as one that falls below (above) the median in total enrollment in the 2006-2007 AY. A low (high) baseline VA school is defined as one that falls below (above) the median in VA in the 2006-2007 AY. A school in a poor (rich) neighborhood is defined as one that is located in a census block group that falls below (above) the median for average income in 2010. A school is said to be nearby a higher quality private schools if the nearest private choice school had a higher baseline overall school VA. The number of observations in Panel E are significantly lower because often the nearest eventual private choice school does not have available testing data in the baseline year. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).

Table 6: The Competitive Effects of ICSP on Private Choice Schools

VARIABLES	(1) School Overall Value-Added	(2) School Overall Value-Added	(3) School Math Value-Added	(4) School Math Value-Added	(5) School Reading Value-Added	(6) School Reading Value-Added
$Post_t \cdot HighExp_s$	-0.016	0.005	-0.017	0.012	-0.020	-0.003
	(0.018)	(0.026)	(0.022)	(0.035)	(0.014)	(0.025)
Interaction with High Nearest Public		-0.026		-0.034		-0.020
School of Lower Quality		(0.032)		(0.042)		(0.027)
Observations	2,148	2,148	2,148	2,148	2,148	2,148
R-squared	0.346	0.346	0.347	0.347	0.371	0.372

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. High exposure is defined as being in the top tercile of the distribution of the number of public schools within five miles. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).

Table 7: Private Choice School Heterogeneity Results

VARIABLES	(1) School Value-Added	(2) School Math Value-Added	(3) School Reading Value-Added
Panel A: Pre-policy Enrollment Decline			
$Post_t \cdot HighExp_s$	0.006	0.016	-0.007
	(0.015)	(0.018)	(0.016)
Interaction with Enrollment	-0.037	-0.053	-0.020
Decline Between 2007-2011	(0.028)	(0.035)	(0.022)
Observations	2,148	2,148	2,148
Panel B: Below Median Tuition Charges			
$Post_t \cdot HighExp_s$	-0.025	-0.026	-0.024
	(0.026)	(0.032)	(0.019)
Interaction with Below	0.020	0.021	0.010
Median Tuition Charges	(0.028)	(0.034)	(0.022)
Observations	2,148	2,148	2,148
Panel C: Non-"A" School Grade in 2007			
$Post_t \cdot HighExp_s$	-0.025	-0.023	-0.027*
	(0.021)	(0.025)	(0.016)
Interaction with Non-"A"	0.039	0.030	0.032
School Grade in 2007	(0.024)	(0.032)	(0.019)
Observations	2,148	2,148	2,148
Panel D: Below Median Baseline Quality			
$Post_t \cdot HighExp_s$	-0.062**	-0.076**	-0.045**
	(0.028)	(0.033)	(0.020)
Interaction with Below	0.086***	0.112***	0.048**
Median Baseline Quality	(0.029)	(0.035)	(0.023)
Observations	2,148	2,148	2,148

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. High exposure is defined as being in the top tercile of the distribution of the number of public schools within five miles. Panel A includes an interaction term with an enrollment decline in the pre-policy years, 2007-2011. Panel B includes an interaction term if the private choice school reported a tuition below the median in its first year accepting voucher students. Panel C includes and interaction term with whether the private choice school received a non-"A" grade from the state in the baseline year. Panel D includes an interaction term indicating whether a choice school was below the median in school value-added in the 2006-2007 academic year. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).

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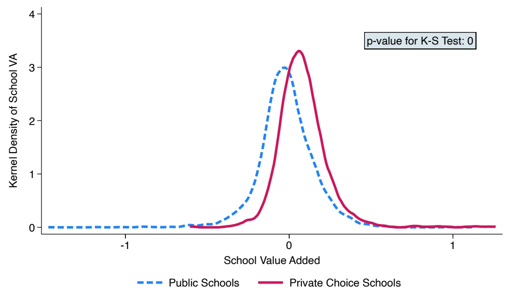
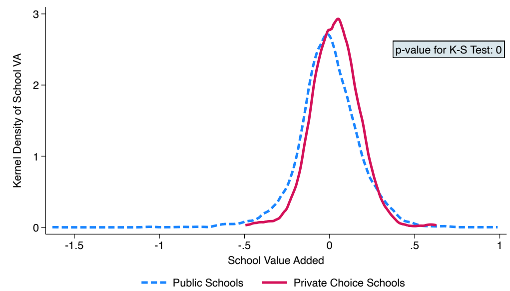
Figure A1: Kernel Density Plots - Public and Private Choice Schools

(a) Pre-Policy



(b) Post-Expansion

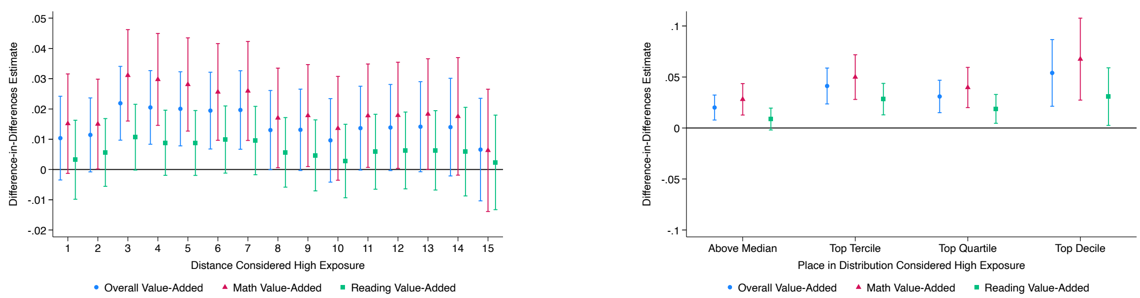


Notes: This figure depicts the kernel density plots of our school value-added (VA) estimates for the public and private choice schools in our sample. Panel A shows the kernel density plots of schools in the years before the voucher program was implemented. Panel B shows those same estimates in the years after the program was expanded. School VA estimates are calculated using the OLS regression described by Equation (1). Data on test scores and enrollment come from the IDOE-CREO database. The p-value for the Kolmogorov-Smirnov equality-of-distributions test is reported.

Figure A2: Sensitivity to Definition of High Exposure for Public Schools

(a) Distance to Nearest Competitor

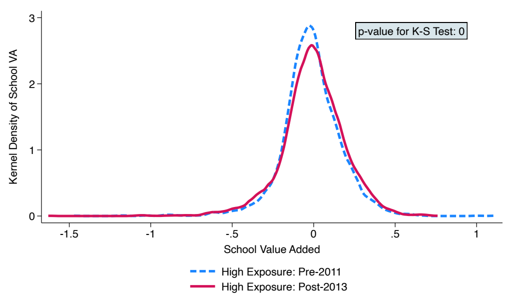
(b) Number of Nearby Competitors



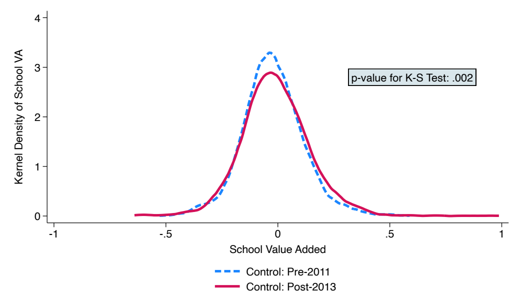
Notes: This figure shows the robustness of our public school results to various definitions of high exposure. Panel A changes the definition of high exposure by varying the distance between a public school and its nearest private choice school competitor required to be considered nearby. Panel B changes the definition by varying the number of nearby competitors public schools have within five miles. Each marker is a result from a separate regression. Blue circles represent the results for overall value-added, red triangles represent the results for math value-added and green circles represent the results for reading value-added. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).

Figure A3: Public and Private Choice School Value-Added Pre- and Post-Policy

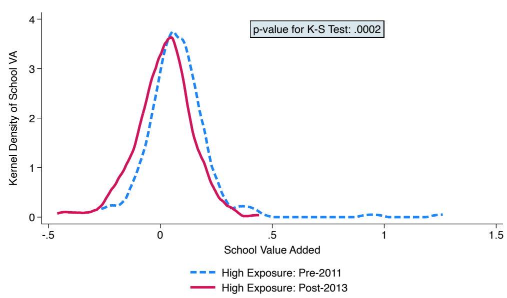
(a) Public High-Exposure Schools



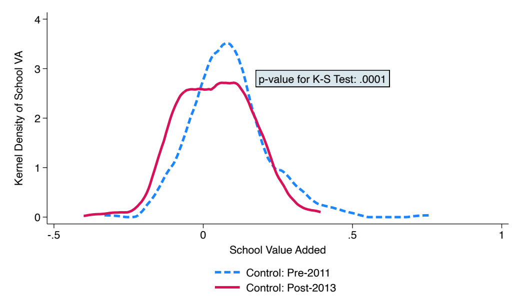
(b) Public Control Schools



(c) Private Choice High-Exposure Schools

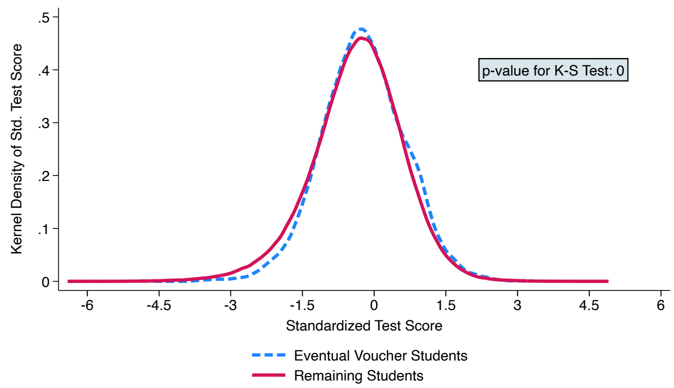


(d) Private Choice Control Schools



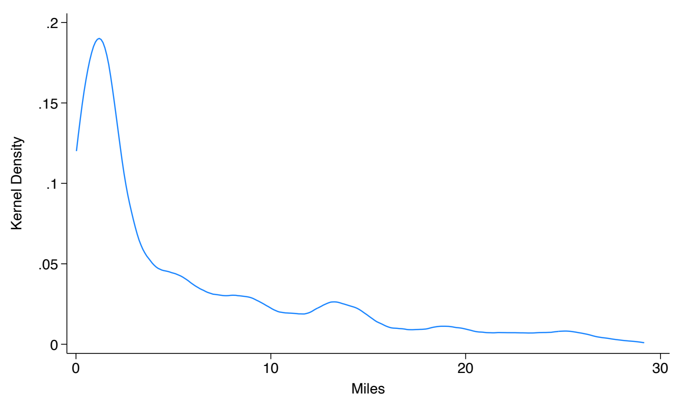
Notes: This figure depicts the kernel density plots of our school value-added (VA) estimates for the public and private choice schools in our sample. Each panel plots school VA across two time periods: pre-2011 and post-2013 to align with the policy time horizons. Panels A and C show the kernel density plots of schools facing high-exposure to the policy. Panels B and D show the kernel density plots for the control group. For public schools, high-exposure is defined as having an eventual private choice school within five miles of the school's location. For private choice schools, high exposure is defined as being in the top tercile of the distribution of the number of public schools within five miles. School VA estimates are calculated using the OLS regression described in Equation (1). Data on test score and enrollment come from the IDOE-CREO database. The p-value for the Kolmogorov-Smirnov equality-of-distributions test is reported.

Figure A4: Kernel Density Plots of Standardized Test Scores



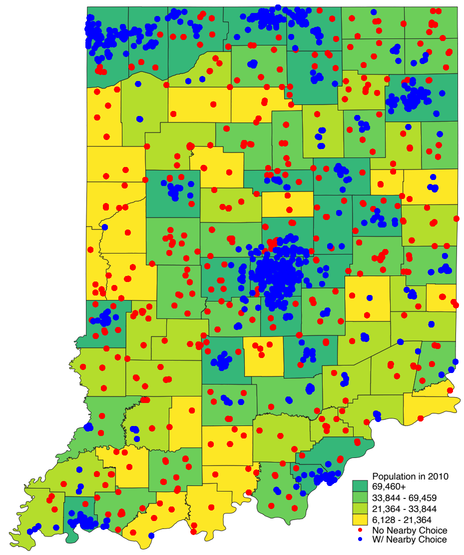
Notes: This figure depicts the kernel density plots of standardized test scores for the students attending public schools in the years before the voucher program was adopted. This figure shows the kernel density plots for the eventual voucher students and students remaining in the public school despite qualifying for a 90% voucher. The p-value for the Kolmogorov-Smirnov equality-of-distributions test is reported.

Figure A5: Kernel Density Plot of Distance to Nearest Private Choice School



Notes: This figure depicts the kernel density plot of the distances between every public school in our sample and the nearest eventual private choice school. Distance is calculated using radial distances between physical addresses. Data on addresses of schools comes from the IDOE-CREO database.

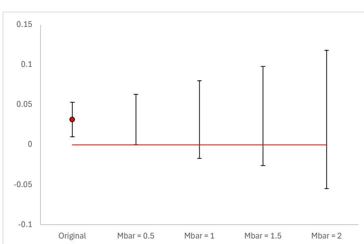
Figure A6: Locations of High Exposure and Control Public Schools



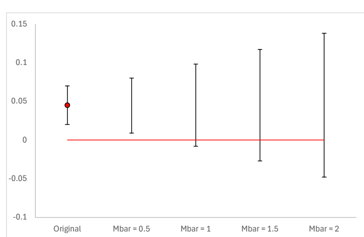
Notes: This figure plots the location of each public school in our sample across Indiana. The blue dots indicate the public schools that have an eventual private choice school within five miles of its location. The red dots represent the public schools in our control group. The map also shows the population counts for each county in the state in the year 2010. Yellow counties are the least populous, while dark green counties are the most populous. Data on the locations of schools comes from IDOE-CREO database and information on population comes from the U.S. Census Bureau, 2010 Census.

Figure A7: Parallel Trends Sensitivity

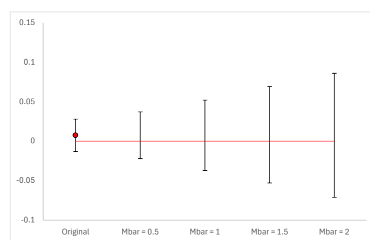
(a) Overall School Value-Added



(b) Math Value-Added



(c) Reading Value-Added

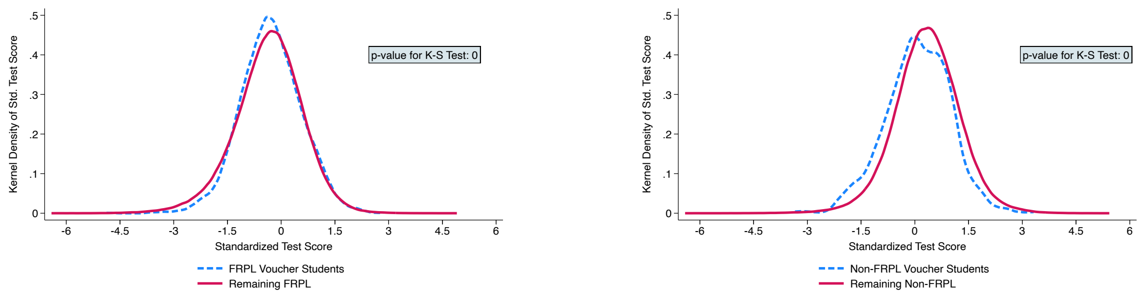


Notes: This figure depicts the results of the event-study sensitivity analysis proposed in Rambachan and Roth (2023). Each panel plots the robust 95% confidence sets for the treatment effect in the year the Indiana Choice Scholarship Program was expanded (2013-2014 academic year) using different values of the maximum post-treatment violation of parallel trends between consecutive periods (Mbar). Panel A shows these results for overall school value-added, Panel B shows these results for math school value-added and Panel C shows these results for reading school value-added. The breakdown value corresponds to the first value of Mbar that results in a confidence set that includes zero, suggesting a statistically insignificant treatment effect.

Figure A8: Student Sorting Across FRPL Status

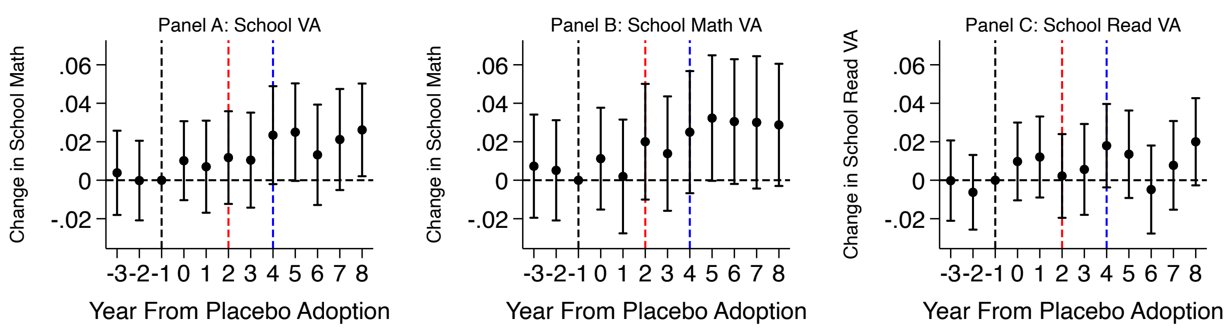
(a) FRPL Voucher vs. FRPL Remain

(b) Non-FRPL Voucher vs. Non-FRPL Remain



Notes: This figure depicts the kernel density plots of standardized test scores for the students attending public schools in the years before the voucher program. Each panel plots test scores for students who eventually use a voucher and those that remain in public school. Panel A shows the kernel density plots for FRPL eventual voucher students and all FRPL remaining public school students. Panel B shows the kernel density plots for non-FRPL eventual voucher students and non-FRPL students remaining in the public school. Test scores are standardized by year and grade. Data on test scores and enrollment come from the IDOE-CREO database. The p-value for the Kolmogorov-Smirnov equality-of-distributions test is reported for each panel.

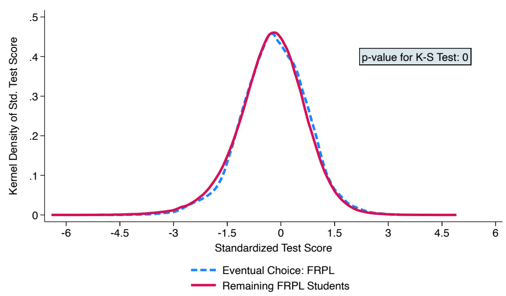
Figure A9: Placebo Event-Study Results for Public Schools



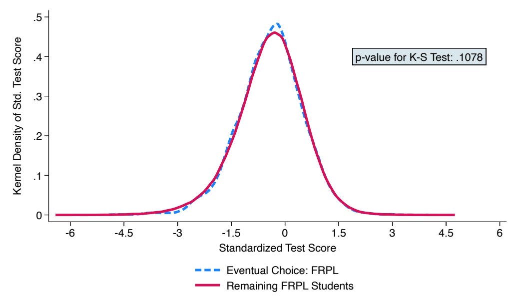
Notes: This figure presents the event-study estimates using placebo treatment years. Figure A9(a) plots the estimates for overall school value-added, Figure A9(b) plots the estimates for school math value-added and Figure A9(c) plots the estimates for school reading value-added. Each figure is the result of a separate estimation. 95% confidence intervals are reported. All regressions include school and year fixed effects. Baseline covariates include the share of students that are female, white, black, section 504, special education, and receive testing accommodations in the 2006-2007 academic year. High exposure is defined as having at least one private choice school within five miles. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).

Figure A10: Student Sorting Across Initial VA and FRPL Status

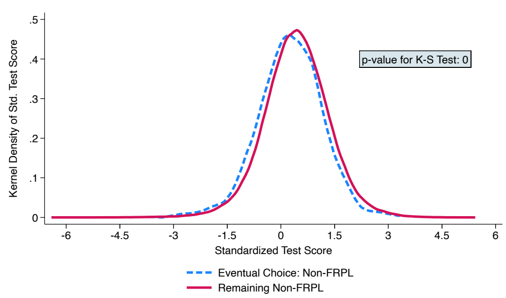
(a) High VA Public School: FRPL



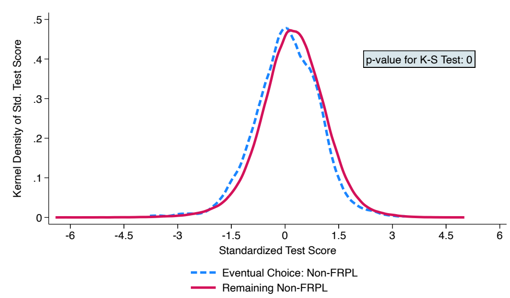
(b) Low VA Public School: FRPL



(c) High VA Public School: Non-FRPL

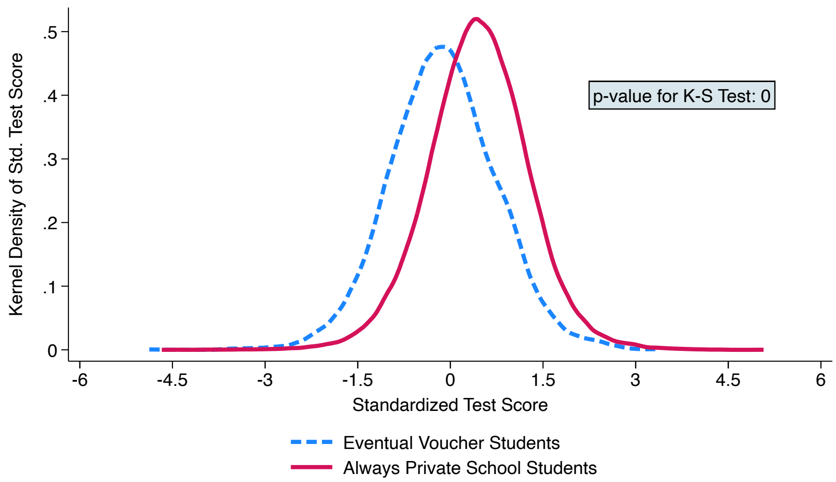


(d) Low VA Public School: Non-FRPL



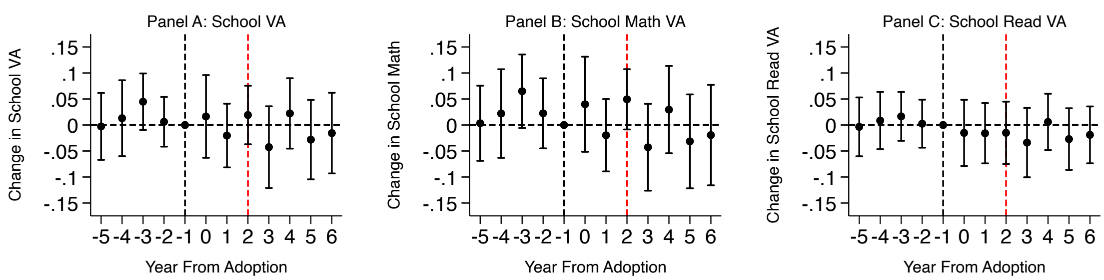
Notes: This figure depicts the kernel density plots of standardized test scores for public school students. Each panel plots test scores across two groups, differentiating between voucher and FRPL status. Panels A and C show the kernel density plots of schools with a baseline overall school quality measure above the median. Panels B and D show the kernel density plots of schools with a baseline overall school quality measure below the median.Data on test score and enrollment come from the IDOE-CREO database. The p-value for the Kolmogorov-Smirnov equality-of-distributions test is reported.

Figure A11: Density Plots of Standardized Test Scores - Voucher vs. Always Private



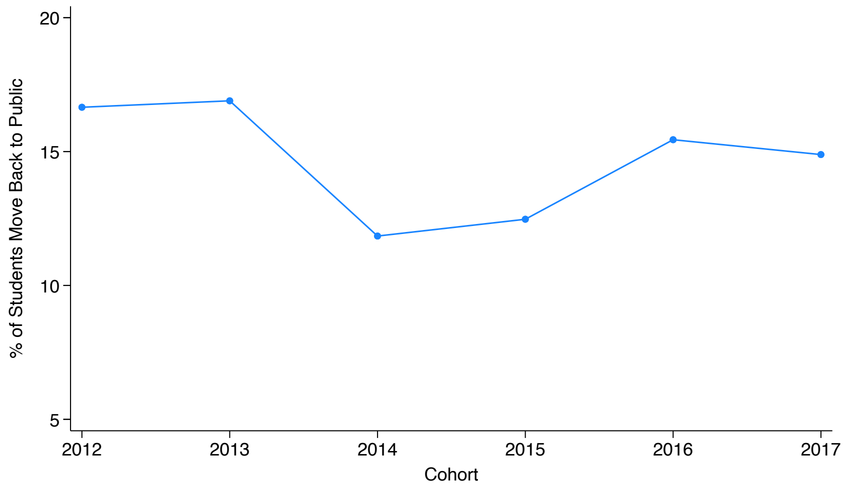
Notes: This figure depicts the kernel density plots of standardized test scores for the students attending a private choice school and had test scores in the years before the voucher program. The figure plots the test scores for students who eventually use a voucher and those that always attended the private school. Test scores are standardized by year and grade. Data on test scores and enrollment come from the IDOE-CREO database. The p-value for the Kolmogorov-Smirnov equality-of-distributions test is reported for each panel.

Figure A12: Private Choice School Event-Study Results



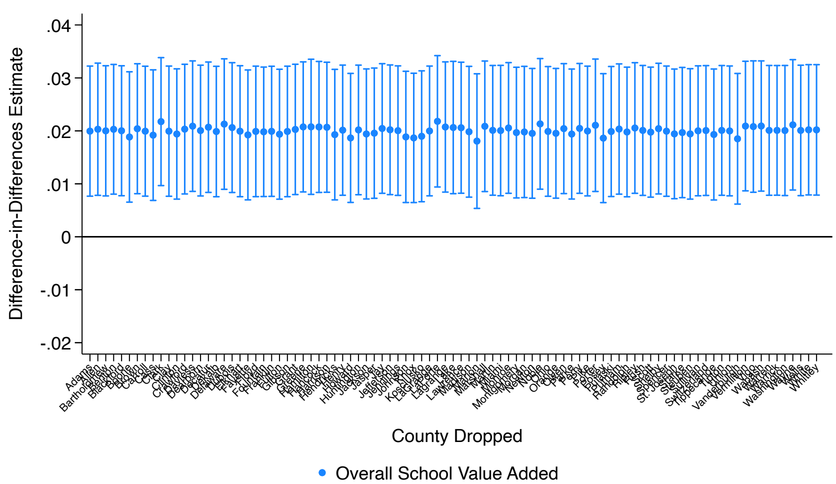
Notes: This figure presents the event-study estimates from Equation (3). Figure 1(a) plots the estimates for overall school value-added, Figure 1(b) plots the estimates for school math value-added and Figure 1(c) plots the estimates for school reading value-added. Each figure is the result of a separate estimation. 95% confidence intervals are reported. All regressions include school and year fixed effects. Baseline covariates include the share of students that are female, white, black, and receive testing accommodations in the 2006-2007 academic year. High-exposure is defined as being in the top tercile of the distribution of the number of nearby public schools. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).

Figure A13: Percent of Voucher Students Moving Back to Public Schools



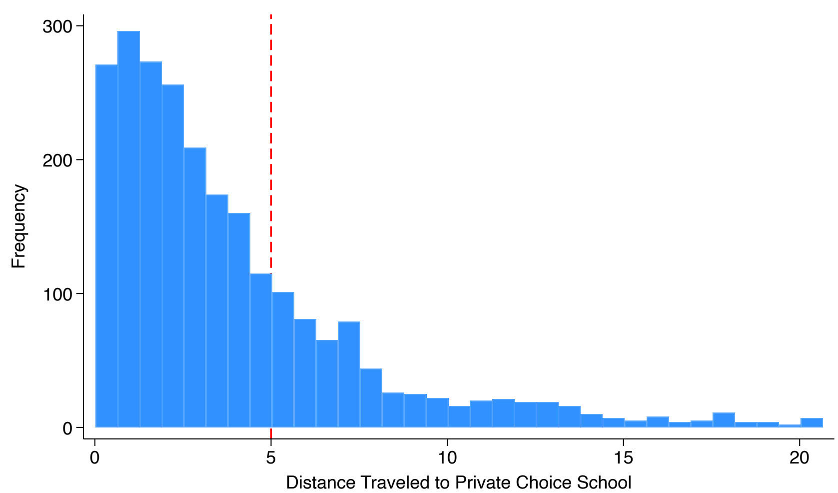
Notes: This figure depicts the percent of voucher students that return back to the public school system in the following year. Data on enrollment come from the IDOE-CREO database.

Figure A14: Public Schools Results Dropping Each County in Indiana



Notes: This figure depicts the results from Equation (2) for our public school sample when we drop the indicated county from the sample. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that are female, white, black, section 504, special education, and receive testing accommodations in the 2006-2007 academic year. High exposure is defined as having at least one nearby choice school. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).

Figure A15: Distance Traveled by Voucher Students



Notes: This figure depicts a histogram of the distance traveled by participating voucher students using the radial distance between the home and attending school address of a random sample of 2,500 students. Data on addresses come from the IDOE-CREO database.

Table A1: DiD Results With Shrunken Value-Added Estimates

VARIABLES	(1) Shrunk School VA	(2) Shrunk Math VA	(3) Shrunk Reading VA
$Post_t \cdot HighExp_s$	0.015***	0.022***	0.005
	(0.005)	(0.007)	(0.004)
Observations	15,201	15,201	15,201
R-squared	0.470	0.443	0.455

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. High exposure is defined as having at least one nearby choice school. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1) and shrunk according (Kane and Staiger, 2008).

Table A2: Summary Statistics of Private Schools In and Out of the Sample

	(1) In Sample	(2) Out of Sample	(3) Difference
# of Students Enrolled	252	94	-158***
	(155)	(168)	
# of Teachers	17	8	-9***
	(10)	(13)	
% White Students	84.17	84.41	0.24
	(21.12)	(24.33)	
% Black Students	3.45	7.68	4.23***
	(9.33)	(18.90)	
% Hispanic Students	7.22	3.87	-3.35**
	(16.02)	(10.05)	
Student-Teacher Ratio	14.22	12.33	-1.89***
	(3.15)	(6.71)	
% Full-Time Teachers	0.88	0.89	.01
	(0.10)	(0.17)	
N	176	533	709

Notes: This table presents summary statistics for the set of schools in and out of the private school sample in the year before the voucher policy was implemented. Data on private schools come the Private School Universe Survey. Three private choice schools in the sample are not found in the Private School Universe Survey.

Table A3: DiD Results Varying School VA Estimation

VARIABLES	(1) Baseline	(2) School-Year FE Only	(3) Including Demographics	(4) Including Previous Test Score
$Post_t \cdot HighExp_s$	0.020***	0.022**	0.024**	0.022***
	(0.006)	(0.011)	(0.011)	(0.006)
Observations	15,228	15,228	15,228	15,228

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. High exposure is defined as having at least one private choice school within five miles. Data on enrollment come from the IDOE-CREO database.

Table A4: Alternative Public School DiD Specifications

VARIABLES	(1) School Value-Added	(2) School Math Value-Added	(3) School Reading Value-Added
Panel A: Continuous Measure of Treatment			
$Post_t \cdot NumClose_s$	0.003***	0.003***	0.002*
	(0.001)	(0.001)	(0.001)
Observations	15,228	15,228	15,228
Panel B: Results Using Control Schools			
$Post_t \cdot HighExp_s$	-0.002	-0.004	-0.001
	(0.010)	(0.012)	(0.009)
Observations	5,808	5,808	5,808
Panel C: Results Removing Schools with Competitor within 3-8 Miles			
$Post_t \cdot HighExp_s$	0.021***	0.028***	0.010
	(0.008)	(0.009)	(0.007)
Observations	12,192	12,192	12,192
Panel D: No Baseline Controls			
$Post_t \cdot HighExp_s$	0.001	0.008	-0.010*
	(0.006)	(0.007)	(0.005)
Observations	15,228	15,228	15,228
Panel E: Removing Marion County			
$Post_t \cdot HighExp_s$	0.018***	0.026***	0.007
	(0.006)	(0.008)	(0.006)
Observations	13,602	13,602	13,602
Panel F: Removing Voucher Peers			
$Post_t \cdot HighExp_s$	0.015	0.027**	0.002
	(0.010)	(0.011)	(0.010)
Observations	9,584	9,584	9,584

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. Panel A uses a continuous measure of the number private choice schools within five miles. Panel B limits the sample to include only public schools in the original control group and designates high exposure as having a private choice school competitor within eight miles. Panel C limits the sample to only include public schools whose nearest private choice school competitor is either less than three miles away or more than eight miles away. Panel D excludes baseline controls. Panel E removes public schools that reside in Marion county. Panel F reports the results when school VA is calculated based only on students that did not attend classes with eventual voucher students. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).

Table A5: Student-Level Results

VARIABLES	(1) Standardized Test Score	(2) Standardized Math Score	(3) Standardized Reading Score
$Post_t \cdot HighExp_s$	0.012**	0.019***	0.001
	(0.005)	(0.006)	(0.004)
Observations	3,742,908	3,742,908	3,742,908
R-squared	0.773	0.705	0.670

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. High exposure is defined as having at least one private choice school within five miles. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).

Table A6: Public Schools Results by Title I Status

VARIABLES	(1) School Value-Added	(2) School Math Value-Added	(3) School Reading Value-Added
Panel A: Had Title I Program			
$Post_t \cdot HighExp_s$	0.021***	0.032***	0.009
	(0.007)	(0.008)	(0.006)
Interaction with Had	0.001	-0.004	0.002
Title I Program in 2010	(0.008)	(0.010)	(0.007)
Observations	15,216	15,216	15,216
R-squared	0.454	0.427	0.445
Panel B: Close to Title I Eligibility Threshold			
$Post_t \cdot HighExp_s$	0.024***	0.034***	0.010*
	(0.007)	(0.008)	(0.006)
Interaction with Close to	-0.010	-0.016	-0.001
Title I Eligibility Threshold	(0.008)	(0.010)	(0.008)
Observations	15,216	15,216	15,216
R-squared	0.455	0.427	0.445

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. High exposure is defined as having at least one private choice school within five miles. Panel A includes an interaction term that indicates whether a high-exposure public school had a Title I program in 2010. Panel B includes an interaction term that indicates whether a high-exposure public school was within 5 p.p. of the cutoff for Title I eligibility. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).

Table A7: Results on Public School Inputs

	(1) Student-Teacher Ratio	(2) # of Teachers w/Graduate Degree	(3) # of HQ Certified Teachers	(4) Average Years of Experience	(5) # of Non-White Teachers	(5) Teacher Retention Rate
$Post_t \cdot HighExp_s$	0.142	0.566**	1.405***	-0.094	0.003	2.117**
	(0.147)	(0.235)	(0.316)	(0.151)	(0.035)	(1.036)
Observations	9,882	10,092	10,092	10,092	9,900	9,862
Baseline Mean	18.21	16.45	21.64	15.32	0.664	79.64

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. High exposure is defined as having at least one private choice school within five miles. Observations are lower compared to other tables because of limited availability of data. Data on teacher characteristics come from the IDOE-CREO database and student-teacher ratios are calculated from the Common Core of Data on Public Schools.

Table A8: Results on Private Choice School Inputs

	(1) Full-Time Teachers	(2) Student-Teacher Ratio	(3) Hours in School Day
$Post_t \cdot HighExp_s$	-0.434	0.833**	-0.069
	(0.844)	(0.406)	(0.052)
Observations	983	983	983
Baseline Mean	15.51	14.22	6.87

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. High-exposure is defined as being in the top tercile of the distribution of the number of public schools within five miles. Data on choice school inputs comes from the Private School Universe Survey which is conducted biannually. There are fewer observations in this analysis because of the survey design.

Table A9: Alternative Private Choice School DiD Specifications

VARIABLES	(1) School Value-Added	(2) School Math Value-Added	(3) School Reading Value-Added
Panel A: Top Quartile of the Distribution			
$Post_t \cdot HighExp_s$	-0.018	-0.019	-0.020
	(0.020)	(0.024)	(0.016)
R-squared	0.346	0.346	0.372
Panel B: Continuous Measure of Number of Public Schools within 5 Miles			
$Post_t \cdot HighExp_s$	-0.001	-0.001	-0.001
	(0.001)	(0.001)	(0.001)
Median Number of Close Public Schools	14	14	14
R-squared	0.346	0.347	0.372
Panel C: No Baseline Controls			
$Post_t \cdot HighExp_s$	-0.009	-0.016	-0.008
	(0.017)	(0.021)	(0.014)
R-squared	0.340	0.341	0.367

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. Panel A defines high exposure as being in the top quartile of the distribution of the number of public schools within five miles. Panel B uses a continuous measure for the number of public schools within five miles. Panel C excludes baseline controls in the specification where high exposure is defined as the being in the top tercile of the distribution of the number of public schools within five miles. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).

Table A10: Alternative High Exposure Definitions for Private Choice Schools

VARIABLES	(1) School Value-Added	(2) School Math Value-Added	(3) School Reading Value-Added
Panel A: Above Median Share of FRPL Public School Students within 5 miles			
$Post_t \cdot HighExp_s$	-0.008	-0.002	-0.021*
	(0.013)	(0.016)	(0.012)
R-squared	0.346	0.347	0.371
Panel B: Within 5 Miles of Initially Low Value-Added Public School			
$Post_t \cdot HighExp_s$	-0.007	-0.003	-0.013
	(0.015)	(0.018)	(0.013)
R-squared	0.346	0.347	0.371

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. Panel A defines high exposure as having an above median share of qualifying public schools students within five miles. Panel B defines high exposure as being within five miles of a public school with overall school value-added below the baseline median. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).

Table A11: DiD Results Relaxing Sample Restrictions

	Public Schools			Private Choice Schools		
	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	School Overall Value-Added	School Math Value-Added	School Reading Value-Added	School Overall Value-Added	School Math Value-Added	School Reading Value-Added
Panel A: Drop Schools Missing Data in the Pre-Period						
$Post_t \cdot HighExp_s$	0.020***	0.028***	0.010*	-0.015	-0.019	-0.015
	(0.006)	(0.008)	(0.005)	(0.017)	(0.021)	(0.013)
Observations	16,038	16,038	16,038	2,247	2,247	2,247
R-squared	0.448	0.420	0.437	0.306	0.336	0.316
Panel B: Keep Schools with Data between 2009-2013						
$Post_t \cdot HighExp_s$	0.019***	0.026***	0.010*	-0.016	-0.020	-0.016
	(0.006)	(0.008)	(0.005)	(0.016)	(0.021)	(0.012)
Observations	15,959	15,959	15,959	2,353	2,353	2,353
R-squared	0.448	0.417	0.439	0.311	0.332	0.333

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. High exposure is defined as having at least one private choice school within five miles. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).

Table A12: Results on Public School Enrollment

	(1)	(2)
VARIABLES	Total # Students All Sample Years	Total # Students Prior to Expansion
$Post_t \cdot HighExp_s$	0.461	-2.280
	(3.326)	(2.827)
Observations	15,228	8,883
Mean of Outcome	246.2	246.2

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. High exposure is defined as having at least one private choice school within five miles. Data on enrollment come from the IDOE-CREO database.

B1 Appendix - Entry and Exit of Schools

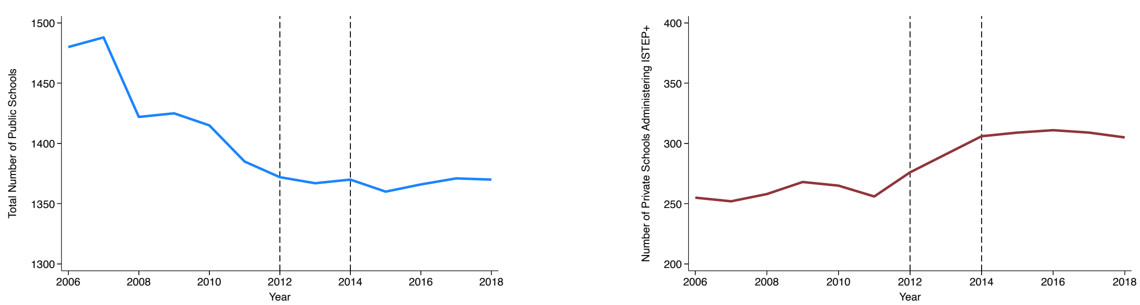
Entry and exit into the market is an important supply-side response to consider when evaluating a voucher program. The goal of this appendix section is to illustrate that the Indiana Choice Scholarship Program did not induce significant entry or exit for either public or private schools.

Number Public and Private Schools

To understand how the number of schools has changed over the sample period, Figure B1 plots the number of public and private schools in each year from 2006-2019, respectively. Using the Common Core of Data, we find at the start of our sample period there are 1,480 public school serving grades 3-8 across the state. That number falls to 1,370 by the end sample. Importantly, there does not seem to be a significant change in the number public schools around policy adoption or expansion. Using the ISTEP+ data, we find at the start of our sample there are 255 private schools across the state serving grades 3-8. That number rises to 305 by the end of the sample period. We do find some evidence that following the adoption of ICSP there was a brief period of time where new private schools started testing their students with ISTEP+. However, we cannot distinguish whether these are new private schools or were existing private schools that started testing students in order to qualify for the voucher program.

Figure B1: Public and Private Schools Across the State

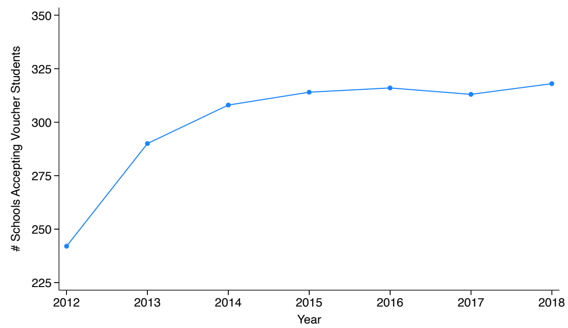
(a) Number of Public Schools (b) Number of Private Schools



Notes: This figure plots the total number of public (Panel A) and private schools using the ISTEP+ exam (Panel B) in the state of Indiana in each year from 2006-2018. Data is shown only for schools that cater to grades 3-8. Data on the number of schools comes from the NCES Common Core of Data and the IDOE-CREO database.

Figure B2 examines the change in the total number of private schools accepting voucher students across our sample period including both primary, secondary and high schools. While there is a steady increase in the number of private schools accepting voucher students in the first couple of years of the program, since the expansion of the program the number of private choice schools has remained steady.

Figure B2: Number of Private Schools Accepting Voucher Students



Notes: This figure plots the total number of private schools in the state of Indiana that accept voucher students for each year from 2012-2018. Data is shown for all schools catering to grades K-12. Data on the number of schools comes from the IDOE-CREO database.

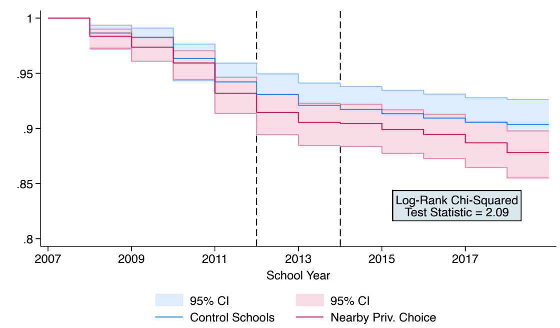
Closures of Traditional Public Schools

We pay particular attention to the closure of traditional public schools because as discussed in Chen and Harris (2022) these events could induce student sorting that biases our results.[35] We investigate this potential mechanism by first examining if public schools located near a private choice school saw a differential increase in their likelihood to close. We estimate a Kaplan-Meier survivor function as shown in Figure B3. It is visually apparent that being located within five miles of a private choice school did not increase the likelihood that the public school would close.

We next examine whether the quality of the closed public schools differed between those with a nearby private choice school and those without one. If the high exposure public schools in our sample receive students from higher quality, closed public schools (when compared to the control group), our estimates may be biased upward. We, therefore, examine whether the distributions of school value-added are equal between these two groups of closed public schools in the years before they close. Figure B4 plots the kernel density functions for the set of public schools that close throughout our sample period. The kernel density functions are estimated separately for the (eventually closed) public schools within five miles of a private choice school and those without a nearby private choice competitor. We fail to reject the null hypothesis of the Kolmogorov-Smirnov equality of distributions test. We take this as suggestive evidence that the quality of the closed public schools did not differ based on the distance to the nearest private choice school.

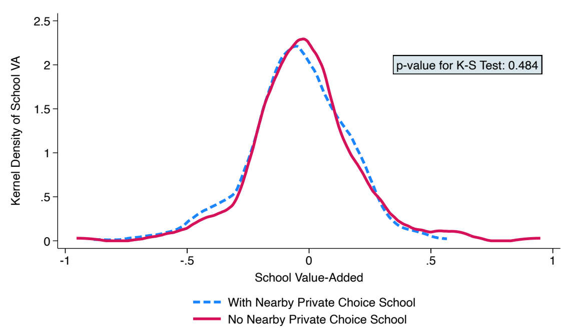
[35] Chen and Harris (2022) explore this idea in the context of charter school penetration.

Figure B3: Survivor Model for Public School Closures



Notes: This figure depicts the Kaplan-Meier survivor function for the set of public schools that existed in Indiana at the start of the sample period. The graphs are separated by whether or not the observation has a private choice school with five miles of its location. The chi-squared test statistic for the log-rank test of equality is reported. The black, dashed lines represent the years the voucher program was implemented and expanded.

Figure B4: Kernel Density Plot of School VA- Closed Schools



Notes: This figure depicts the kernel density plots of our school value-added (VA) estimates for the set of public schools that closed during the sample period. The blue, dashed line shows the kernel density plot for the public schools that were within five miles of a private school. The red, solid line shows the same estimate for public schools farther from a private choice school. School VA estimates are calculated using the OLS regression described in Equation (1). Data on test scores and enrollment come from the IDOE-CREO database. The p-value for the Kolmogorov-Smirnov equality-of-distributions test is reported.

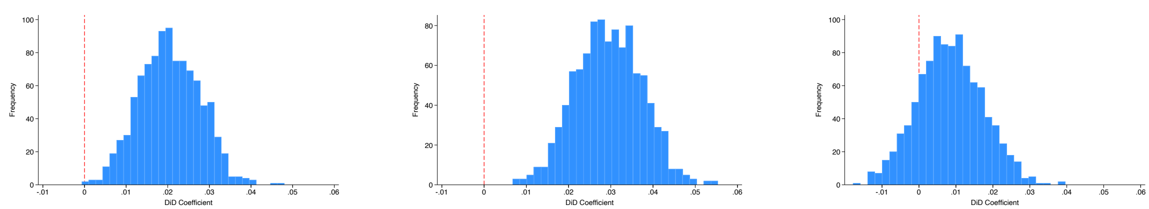
C1 Appendix - Bootstrapping

Deeb (2021) shows that when value-added is the outcome variable of interest in a regression, the regression's robust standard errors used to draw inference are invalid. We, therefore, propose a bootstrapping procedure to correct this issue for our public school analysis.

In each of 1000 iterations, we sample 100 students (with replacement) within each school to be included in the value-added regression described in Equation (1). Therefore, each iteration returns a unique set of school value-added measures that we can use in our difference-in-differences specification. Given this set up, we run Equation 2 on each set of unique school value-added estimates and plot the results as a histogram. We construct new confidence intervals using the standard error of this distribution of difference-in-differences results. We perform this exercise separately for overall school value-added, math school value-added and reading school value-added.

Figure C1 depicts the results of this exercise. Each panel shows the histogram of difference-in-differences results for each of our outcome variables of interest. We highlight where in the distribution the coefficient equals zero to give a sense of the number of iterations that resulted in the voucher program having zero effect on high-exposure public schools. Across all of the panels, it is evident that a majority (if not all) iterations resulted in positive effect of the program on high-exposure public schools. Table C1 displays the average difference-in-differences coefficient along with the newly constructed confidence intervals.

Figure C1: Bootstrapped DiD Results



(a) School Overall VA (b) School Math VA (c) School Reading VA

Notes: This figure depicts the histogram of results from our bootstrapping exercise. Panel A shows the results on Overall School Value-Added (VA), Panel B shows the results on School Math VA, and Panel C shows the results on School Reading VA. A description of the bootstapping exercise can be found in Appendix C1.

Table C1: DiD Results with Bootstrapped Confidence Intervals

VARIABLES	(1) School Overall Value-Added	(2) School Math Value-Added	(3) School Reading Value-Added
Average DiD Estimate	0.020	0.029	0.008
Bootstrapped 95% CI	[0.0199, 0.0209]	[0.0289, 0.0299]	[0.0078, 0.0089]

Notes: Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. High exposure is defined as having at least one private choice school within five miles. Confidence intervals are calculated according to the procedure describe in C1.

D1 Appendix - Predicted Value-Added Exercise

To understand the extent that documented changes in student demographics drive our results, we perform the following exercise using predicted school value-added for both the public and private choice school samples.

Specifically, we begin by estimating the following model:

$$VA_{s,2007} = \sigma X_s^{2007} + \epsilon_s \tag{4}$$

where $VA_{s,2007}$ is our estimated school value-added in 2007 (our "base" year), and X_s^{2007} includes all of the school characteristics we observe (including average test score in prior year) and their pairwise interactions in that same year. We use the coefficients from this fully interacted model to predict value-added for each school in all years of the sample. We then use these predicted value-added measure to run the following difference-in-differences specification:

$$V\hat{A}_{st} = \beta_1 Post_t \cdot HighExposure_s + \delta_s + \gamma_t + \epsilon_{st} \tag{5}$$

If changes in observable school characteristics are driving our school quality results, we would expect differential changes in the predicted value-added measures following the implementation of the voucher policy. Appendix Table D1 reports the results of this exercise for the public school (Panel A) and private choice school (Panel B) samples. We find no evidence that high exposure public schools were predicted to improve their quality based on the change in composition of their students. In fact, we find that based solely on changes in observable characteristics, high exposure public schools were predicted to see declines in all of our outcome variables of interest. We take this result as strong evidence that it is changes made by schools that drive the improvements in quality we see.

For private choice schools, we find that based solely on changes in observable characteristics, high exposure private choice schools were predicted to see statistically insignificant declines in overall value-added, increases in math value-added and no changes in reading value-added. These results show that it is possible that changes in student composition play a role in the results we originally reported for private choice schools. However, we do want to highlight that our results also point to changes made by the choice schools that could explain the declines in quality, specifically Figure A8 shows an increase in student-teacher ratios.

We also recognize that this exercise can only speak to how changes in observable school characteristics may have affected our school-quality results. The concern remains that non-random student sorting on unobservable characteristics is driving our results.

Table D1: DiD Results on Predicted School Value-Added

VARIABLES	(1) Predicted School VA	(2) Predicted School Math VA	(3) Predicted School Reading VA
Panel A: Public Schools			
$Post_t \cdot HighExp_s$	-0.019***	-0.010**	-0.026***
	(0.004)	(0.004)	(0.004)
Observations	15,228	15,228	15,228
R-squared	0.639	0.566	0.674
Panel B: Private Choice Schools			
$Post_t \cdot HighExp_s$	-0.031	0.208	0.005
	(0.091)	(0.129)	(0.154)
Observations	2,148	2,148	2,148
R-squared	0.190	0.161	0.205

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. Regressions include school and year fixed effects. High exposure is defined as having at least one nearby private choice school. Data on test scores come from the IDOE-CREO database. Predicted School Value-Added are estimated by regressing value-added in 2007 on school characteristics and using the regression coefficients to predict school-value added for all years in the sample.

E1 Appendix - Private School Selection into ICSP

For the public school sample, high exposure is defined by whether the school is within five miles of an eventual private choice school. One potential concern with this definition is that private school participation into the program is endogenous to nearby public school quality, potentially biasing our results. For example, private schools may attempt to maximize demand for their services by participating in the program if they are nearby public schools where performance is declining, which might produce a spurious positive relationship through mean reversion. Alternatively, private schools may be more likely to participate if public school performance is increasing, which would produce a spurious negative relationship for similar reasons.

In this section, we present three reasons why, in the context of the Indiana Choice Scholarship Program (ICSP), it was unlikely that private schools decided to participate in the program based on the quality of surrounding public schools. First, participation in the program by private schools is very high. Of the schools that were accredited (and therefore, immediately able to participate) in the year before ICSP was adopted , 85% became a voucher accepting school by the end of our sample period. Second, anecdotal evidence from interviews with officials at participating schools suggests that private schools often joined the program to help church finances (Turner et al., 2017) and work on the voucher programs in Wisconsin bolsters this claim (Hungerman et al., 2019). Third, we empirically tested whether we could predict private school participation with nearby public school quality and found no relationship.

Specifically, to understand the factors that predict private school participation, we ran the following regression on the set of accredited private schools in 2011:

$$Choiceby2018_s = \alpha + X_{s,2011}^{'}\beta + Z_{p,2011}^{'}\gamma + \phi AvgPublicSchoolQuality_{p,2011} + \epsilon_s$$

where $ChoiceBy2018_s$ is an indicator for whether school, s, was a private choice school by 2018, $X_{s,2011}^{'}$ is a vector of the private school's characteristics in 2011 (the year before the program began) including the share of enrollment that are female, Black, White, Hispanic, qualified for subsidized lunch, received math testing accommodations, received reading testing accommodations, an indicator for witnessing enrollment declines between 2007 and 2011, an indicator for becoming accredited after 2007, an indicator if the private school was Catholic, the radial distance to the nearest public school (in miles) and the median household income for the school's zip code, $Z_{p,2011}^{'}$ is a vector of characteristics for the public schools nearby private school, s, including the average share of enrollment that are female, Black, White, Hispanic, qualified for subsidized lunch, received math testing accommodations, and received reading testing accommodations. We define nearby public schools as either those within five miles of the private school's location (Columns (1)-(3) in the table below) or the closest three public schools (Columns (4)-(6)). We include both definitions because not all accredited private schools in 2011 had a public school within five miles. $AvgPublicSchoolQuality_{p,2011}$ is defined in one of three manners: (a) the average overall school value-added in 2011 in nearby public schools, (b) an indicator for whether the average overall school value-added in the nearby public schools was in the bottom quartile of the distribution, or (c) an indicator for whether the average overall school value-added in the nearby public schools was in the top quartile of the distribution.

Table E1 reports the results of this exercise. Across the various specifications, we do not find strong evidence that surrounding public school quality predicts private school participation into the Indiana Choice Scholarship Program. This finding is robust to both the definition of nearby public school and how we measure public school quality. We understand that this exercise may not fully address the endogeneity concern, but we take it as useful evidence our results are not biased due to the choice of private schools to participate in the program.

In an additional specification, we use information on open, accredited private schools (those we have data for) prior to the adoption of ICSP to redefine our high exposure definition for public schools to whether there exists an accredited private school within five miles of the school's location. Table E2 shows the results of this exercise. We find that our results remain positive and statistically significant (except for the results on reading value-added). The fact that these estimates are slightly smaller than those presented in our main specification is in line with the idea that those newly "treated" under this specification were originally in the control group that did not see meaningful changes in our outcome variables of interest as shown in Panel B of Appendix Table A4.

Table E1: Private School Participation in ICSP

	Using All Public Schools within 5 Miles			Using Closest 3 Public Schools		
Variables	(1) Becomes Private Choice School	(2) Becomes Private Choice School	(3) Becomes Private Choice School	(4) Becomes Private Choice School	(5) Becomes Private Choice School	(6) Becomes Private Choice School
Decline in Enrollment from 2007 to 2011	-0.060	-0.060	-0.058	-0.043	-0.036	-0.038
	(0.046)	(0.045)	(0.045)	(0.045)	(0.044)	(0.044)
Accredited After 2007	0.006	0.006	0.007	-0.002	0.005	0.009
	(0.071)	(0.072)	(0.073)	(0.072)	(0.071)	(0.072)
Catholic Private School	0.050	0.050	0.051	0.047	0.050	0.050
	(0.052)	(0.052)	(0.053)	(0.051)	(0.051)	(0.051)
Distance to Nearest Public School	-0.025	-0.025	-0.025	-0.043*	-0.044**	-0.043*
	(0.029)	(0.030)	(0.030)	(0.022)	(0.022)	(0.022)
Median Household Income in '000s (Zipcode)	0.001	0.001	0.001	-0.001	-0.001	-0.001
	(0.002)	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)
Average Nearby Public School Overall VA in 2011	-0.038			-0.272		
	(0.300)			(0.168)		
Nearby Public Schools in Bottom Quartile		0.009			0.046	
		(0.046)			(0.047)	
Nearby Public Schools in Top Quartile			0.011			-0.040
			(0.044)			(0.052)
Observations	246	246	246	253	253	253
R-squared	0.246	0.246	0.246	0.271	0.267	0.267
2011 Private School Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
2011 Public School Characteristics	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Each coefficient is the result of a separate estimation.

Table E2: Public School DiD Results with Alternate Definition of High Exposure

	(1) School Value-Added	(2) School Math Value-Added	(3) School Reading Value-Added	(4) Percent Days Attended	(5) Total Days Attended	(6) Percent Expelled or Suspended
$Post_t \cdot HighExp_s$	0.014**	0.017**	0.006	0.362***	0.548***	0.000
	(0.006)	(0.008)	(0.006)	(0.101)	(0.188)	(0.002)
Observations	15,228	15,228	15,228	15,216	15,216	15,216
R-squared	0.453	0.425	0.443	0.844	0.836	0.773

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. All standard errors are clustered at the school level. Each coefficient is the result of a separate estimation. All regressions include school and year fixed effects. Baseline covariates include the share of students that identify as female, White, Black and receive testing accommodations in the 2006-2007 academic year. High exposure is defined as having at least one accredited private school within five miles in the 2010-2011 academic year. Data on enrollment come from the IDOE-CREO database and school value-added is calculated using Equation (1).